

Complete-Current Estimates for Coherent Quantum Sources and Continua

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Abstract

We prove quantitative response estimates that retain coherent quantum sources, complete physical currents, spectral poles and actual feedback. An ordered inverse lift converts temporally regular form-dual forcing into a first-form response under common-form work bounds. A finite spectral-window theorem controls varying Hilbert-space inputs, and ordered closed-form resolvent identities transfer noncommuting physical-energy weights. Far-frequency and damped estimates retain the endpoint and source-graph costs. Complementary force, invariant and temporal-domain estimates distinguish valid current resources for singular interactions. In a compact-profile electron-oscillator model, we derive the full Coulomb forcing measure and an actual-source low-window sandwich below 4.05×10^{-9} . A separate finite-time calculation bounds the electron and source current differences by .046 and .004 in their respective coordinate lengths. The spectral supremum is taken over the actual spectrum; a spectral floor supplies only an upper bound. Full-source forcing compression and causal feedback are developed in appendices. These results provide explicit analytic inputs to physical transport and record problems. Event probabilities and a realized measurement apparatus require additional flow, preparation and decoder premises.

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1 Introduction

An error estimate for a quantum amplitude becomes a physical transport estimate only after the relevant derivatives, source coordinates and current constitution have been retained. This distinction is especially important for a coherently driven continuum. A small omitted occupation does not eliminate its interference current; a small omitted forcing does not eliminate response generated by feedback; and an accurately resolved electronic transition does not determine the current of a quantum source. These are analytic questions even before an experiment or a measurement record is specified.

We develop a common set of quantitative interfaces for these questions. The complete wave remains a vector on all admitted positions and internal degrees of freedom. Its error is controlled in a form that dominates the physical kinetic derivatives, then converted to the full coherent current. The response may be driven by an unbounded coupling, contain genuine bound poles and a continuum, and receive a time-dependent vector input from another interacting block. We retain the input of the coupled evolution throughout.

There are three main results. First, an ordered inverse lift converts temporally regular form-dual forcing into a first-form response under absolutely continuous common-form work bounds. It avoids a weighted inverse-square-root derivative that is not uniformly controlled in dimension. Second, an operator spectral-window estimate controls a varying Hilbert-space input. The proof uses interval means, spectral derivatives and orthogonal output sectors; it does not replace the vector input by a scalar supremum. Ordered resolvent identities then transfer genuinely noncommuting physical-energy weights, while explicit far-frequency dressings retain both endpoints. Third, a compact-profile electron-oscillator model supplies an analytic Coulomb spectral calculation, an actual-source inverse bound and a complete finite-time current calculation with source feedback.

Several complementary results make these interfaces usable. A spatial force estimate retains covariant strain, curvature and first-order transport. A temporal common-domain estimate instead accepts a residual without a spatial form norm and retains its right clock. An exact positive quadratic invariant supplies a comparison form even through an inverted trap interval. The appendices preserve a spatial forcing compression with arbitrary source coefficients and the causal high-sector feedback that such compression must not erase. Each has its own input hypotheses; choosing the smallest coefficient from incompatible estimates would not be a valid composition.

1.1 Relation to established methods and prior work

The functional-analytic ingredients have substantial precedents. Kato's smoothing and factorized-resolvent framework [6] and its inhomogeneous development [7] motivate the relation between response and resolvents. Our fixed positive imaginary part is an earned finite-resolution coefficient, not a claim of a uniform limiting-absorption estimate. Common-form propagation belongs to the classical evolution theory of Kiszyński [8]; we give the argument for the precise absolutely continuous relative-work assumptions used here. Common *operator*-domain regularity, used in the separate temporal graph theorem, is discussed in [9]. We keep these two kinds of domains distinct.

The oscillator invariant is classical [12, 13]; its contribution here is an explicit physical-current metric with the centroid and mass factors preserved. The Coulomb functions and their normalization are standard [10]; the spectral calculation is included to make the forcing measure and source-weighted bounds reproducible. Isolated-band constructions such as [11] require their own symbol classes and gap hypotheses. Neither a finite column corrector nor a frozen internal gap is asserted here to supply such a band reduction of the complete position/source parent.

The current October research editions of the measurement programme include equilibrium and

nonequilibrium record constructions [4, 5]. Those works ask which original laws yield outcome weights and whether a distinct receiver preserves an earlier actual event. Their apparatus constants and preparation premises are not inputs to the estimates proved here. The new companion manuscripts treat retained-archive preparation (P1) [1], effective repeated records (P2) [2], and reference-compatible flows and whole-history stability (P3) [3]. The present response and current estimates enter such arguments only when their model, entrance and flow premises agree. The proofs below are independent of those companions; the specific flow application in Section 11.2 uses the revised P3 theorem explicitly.

These new manuscripts remain distinct from both the current October website editions and the preserved September editions. No integration into one apparatus is asserted. In particular, assumptions of the pilot-medium and massive-configuration constitutions are not interchangeable. The contribution here is the proved combination of response, derivative and complete-current estimates under stated hypotheses, rather than a priority claim for their established individual ingredients.

1.2 Conventions and the structure of the results

We use $i\partial_t\Psi = H\Psi$ with H in rate units, unless reduced-mass atomic units are explicitly declared. Hilbert-space norms include every physical source coordinate; a finite passive internal reference can be included by tensoring the identity. A source wave or its quantum covariance is not an assumption about an actual configuration law. Form inequalities always refer to the complete stated parent.

Section 2 first supplies the current and coordinate conversion. Sections 3–7 prove the form-dual, spectral-window and ordered-resolvent results. Section 8 develops the alternative spatial-force, invariant and temporal-domain estimates. Sections 9–11 give the hydrogenic example. Appendices A and B preserve the full-source spatial compression and feedback mechanisms.

The hydrogenic example has $\Omega = .01$, $|g| \leq 10^{-6}$ and a radius-40 compact dipole profile in its specified units. Its low-energy physical resolvent sandwich is below 4.05×10^{-9} under the stated source weight. A separate all-frequency, stronger-source-weight estimate gives, for the declared entrance and $T = 100$, integrated electron and source *current differences* below .046 and .004 in their respective coordinate lengths. The comparison uses the actual projected wave, including feedback. It is not a second independently flowing wave, and these numbers are not probabilities.

2 From a coherent error to the complete physical current

We first state the conversion that every response estimate in this paper must supply. All positions, including source positions, belong to $x \in \mathbb{R}^d$. A finite internal fibre is allowed. A wave can also be a matrix of finitely many input columns; all norms below are the spatial L^2 Hilbert–Schmidt norms in that case. This convention gives one envelope before contraction with an unknown input, rather than a sum of incoherent branch currents. Write $\text{Im}_H Z = (Z - Z^\dagger)/(2i)$.

In rate units, consider a self-adjoint local parent with

$$H = - \sum_i D_i \lambda_i D_i + U + V - i \sum_i \left(a_i D_i + \frac{1}{2} \nabla_i a_i \right), \quad D_i = \partial_i - i C_i, \quad \nabla_i X = \partial_i X - i [C_i, X]. \quad (2.1)$$

Here C_i, a_i, V are Hermitian matrices, U is real scalar, and $\lambda_i > 0$ is independent of position; dependence on time is allowed. Domains and coefficient regularity sufficient for the indicated continuity equation are part of this parent. Its density and current are

$$\rho_\Psi = \Psi^\dagger \Psi, \quad J_i[\Psi] = 2\lambda_i \text{Im}_H(\Psi^\dagger D_i \Psi) + \Psi^\dagger a_i \Psi. \quad (2.2)$$

The complete continuity equation is $\partial_t \rho_\Psi + \sum_i \partial_i J_i[\Psi] = 0$; it also holds entrywise for the matrix of input columns evolved by the same parent. Indeed, on a common local core, $\partial_t(\Psi^\dagger \Psi) = i(H\Psi)^\dagger \Psi - i\Psi^\dagger H\Psi$. The Hermitian multiplication terms cancel. The covariant product rule turns the kinetic contribution into $-\sum_i \partial_i [2\lambda_i \operatorname{Im}_H(\Psi^\dagger D_i \Psi)]$ and the symmetrized first-order contribution into $-\sum_i \partial_i (\Psi^\dagger a_i \Psi)$. Testing and passage in the admitted form domain give the weak identity. Thus the first-order term contributes a literal transport current; omitting it would change the conservation law. Additional microscopic spin-curl or nonlocal current constitutions require their own terms; they are not consequences of (2.1).

Proposition 2.1 (Complete coherent current conversion). *Let $\Psi = P + E$ use the same derivative and current constitution, and let a positive form K satisfy $K \geq \sum_i \lambda_i D_i^\dagger D_i$. Put*

$$e = \|E\|, \quad Q = \|K^{1/2}E\|, \quad N = \|\Psi\|, \quad \tau_i = \|\sqrt{\lambda_i} D_i P\|.$$

Whenever the displayed moments are finite,

$$\int \|\rho_\Psi - \rho_P\|_{\text{op}} \leq e(\|P\| + N), \quad (2.3)$$

$$\int \|J_i[\Psi] - J_i[P]\|_{\text{op}} \leq 2\sqrt{\lambda_i}(e\tau_i + NQ) + e(\|a_i P\| + \|a_i \Psi\|). \quad (2.4)$$

If a_i is bounded, the last term is at most $\|a_i\|_\infty e(\|P\| + N)$. Neither normalization nor an independent flow for the comparison P is required.

Proof. Before any spatial integration, the exact identities are

$$\rho_\Psi - \rho_P = E^\dagger P + \Psi^\dagger E$$

and

$$\begin{aligned} J_i[\Psi] - J_i[P] &= 2\lambda_i \operatorname{Im}_H (E^\dagger D_i P + \Psi^\dagger D_i E) \\ &\quad + E^\dagger a_i P + (a_i \Psi)^\dagger E. \end{aligned}$$

The operator norm of a product is bounded by the product of its Hilbert–Schmidt norms, and $\|\operatorname{Im}_H Z\|_{\text{op}} \leq \|Z\|_{\text{op}}$. Spatial Cauchy–Schwarz and $\sqrt{\lambda_i} \|D_i E\| \leq Q$ give the claims. Hermiticity moves the unbounded a_i in the last term onto Ψ , so no unproved bounded-operator coefficient or moment of $a_i E$ has been inserted. $\square$

For a charged particle with physical momentum $p_{\text{ph}} = -i\hbar\partial$, dividing its Hamiltonian by $\hbar$ gives $\lambda_i = \hbar/(2M_i)$ and $C_i = e_i A_i/\hbar$. A symmetrized mechanical term $\{p_{\text{ph}}, d \times B\}/(2M\hbar)$ in the rate generator has $a = (d \times B)/M$: there is no additional factor $1/\hbar$ in its velocity. An unbounded dipole or source operator must be handled by the moments in (2.4).

Example 2.2 (Small wave error with unbounded current error). On the circle of length 2π , let $\Psi = (2\pi)^{-1/2}$ and

$$P_n(t, x) = \frac{1 + n^{-1} e^{in^2 x - i\lambda n^4 t}}{\sqrt{2\pi(1 + n^{-2})}}, \quad n \geq 2.$$

Both are normalized solutions of $i\partial_t u = -\lambda\partial_x^2 u$, and $\|P_n - \Psi\| \leq 2/n$. Nevertheless

$$J[P_n] = \frac{2\lambda}{2\pi(1 + n^{-2})} (n \cos(n^2 x - \lambda n^4 t) + 1),$$

so $\|J[P_n] - J[\Psi]\|_{L^1} \geq 2\lambda(2n/\pi - 1)/(1 + n^{-2}) \rightarrow \infty$. The missing resource is derivative control, not a sharper wave-norm inequality. This example even uses two legitimate free waves.

2.1 Coordinates, moving tests and the original law

Proposition 2.3 (Current under a moving coordinate chart). *Let $(t, x) \mapsto \chi_t(x)$ be jointly C^1 , with each $q = \chi_t(x)$ an orientation-preserving diffeomorphism. Put $G = D_x \chi_t$ and $g = \det G > 0$. A locally integrable laboratory density/current pair (ρ, j) satisfying the continuity equation transforms to*

$$\tilde{\rho} = g \rho \circ \chi_t, \quad \tilde{j} = g G^{-1} (j \circ \chi_t - (\rho \circ \chi_t) \partial_t \chi_t). \quad (2.5)$$

These obey the transformed continuity equation. In particular a moving surface $F(t, q) = n(t) \cdot (q - C(t)) - s$, $|n| = 1$, has relative normal flux

$$j \cdot n + \rho n' \cdot (q - C) - \rho n \cdot C'. \quad (2.6)$$

Proof. For a compactly supported smooth test $\varphi(t, x)$, set $\psi(t, q) = \varphi(t, \chi_t^{-1}(q))$. The chain rule gives $\nabla_q \psi = G^{-T} \nabla_x \varphi$ and $\psi_t = \varphi_t - \nabla_x \varphi \cdot G^{-1} \chi'_t$. Insert these in the weak continuity equation and use $dq = g dx$. This proves (2.5) without a componentwise guess at the current. The derivative along a laboratory path is $dF/dt = F_t + (j/\rho) \cdot \nabla F$; since $|\nabla F| = 1$, its density-weighted normal rate is (2.6). $\square$

The chart formula fixes the transport part of a transformed current. A position-dependent internal unitary also transforms the connection; one must conjugate the whole parent and include that derivative. Integrating hidden source positions first is generally insufficient: $|\int j dy| \leq \int |j| dy$ may discard counterflow or coherent terms needed by a trajectory test.

To state the event interface precisely, suppose the complete current already admits a reference path law with one-time density ρ and velocity j/ρ , and the original actual law satisfies $\mu_0 \leq C\rho_0$. Reweight that same path law by its entrance density ratio. Admit tests h for which $t \mapsto h(t, X_t)$ is absolutely continuous on almost every reference path, with derivative $h_t + (j/\rho) \cdot \nabla h$, and for which the integral below is finite. For example, C^1 tests with this integrability have the required chain rule. Then

$$\mathbb{E}_{\mu_0} \text{Var}_{[0,T]} h(t, X_t) \leq C \int_0^T \int |\rho h_t + j \cdot \nabla h| dq dt. \quad (2.7)$$

Indeed, the chain rule along reference paths, entrance domination and Tonelli's theorem prove this in that order. If $h_t + b_{\text{ref}} \cdot \nabla h = 0$, the integrand becomes $|\nabla h \cdot (j - b_{\text{ref}} \rho)|$. Comparison to $j_{\text{ref}} = b_{\text{ref}} \rho_{\text{ref}}$ therefore costs both $j - j_{\text{ref}}$ and $b_{\text{ref}}(\rho_{\text{ref}} - \rho)$. An event estimate additionally needs the test's guard width, initial bad mass, interval, decoder and own reference crossings. Flow existence and reference-compatible selection are separate premises, studied in the companion flow paper [3]. Equations (2.4) and (2.7) do not assign a new actual law at an intermediate time. The domination premise concerns the complete original joint law, including retained source and archive coordinates. Absolute continuity alone supplies no finite value of C , and an averaged conditional bound supplies no uniform guarantee after arbitrarily rare postselection.

A sharp surface requires an appropriate trace and crossing area formula; a volume L^1 current estimate or an almost-every-level coarea statement does not by itself control flux at a prescribed surface. Smooth guard tests can instead bound a traversal that changes h by a stated positive amount. Tangential contacts and moving guards must satisfy their own chain-rule or crossing premises. All such estimates concern the full time interval in (2.7).

3 Temporal dressing in the physical form dual

Response estimates must keep the input that actually drives the complement. For an admitted fixed block decomposition $\mathcal{H}_P \oplus \mathcal{H}_Q$, the equation

$$i\dot{q} = Aq + Lp$$

uses the actual $p = P\Psi$, including its feedback through q , source phases and clock dynamics. An incident population or a frozen electronic amplitude is not a substitute for this vector. Throughout this section energies are in the units of $i\dot{q} = Aq + f$; a physical Hamiltonian must be divided by $\hbar$ if time is measured in seconds.

Let $\mathcal{V} \subset \mathcal{H} \subset \mathcal{V}^*$ be a dense separable Hilbert form triple, with the usual pivot identification. A positive form $K \geq I$ identifies $\mathcal{V}$ with $\text{Dom } K^{1/2}$ and gives the dual norm $\|f\|_{\mathcal{V}^*, K} = \|K^{-1/2}f\|_{\mathcal{H}}$. Here the square root on a dual vector is its continuous extension, rather than an assertion that $f \in \mathcal{H}$. We use the same letter for a self-adjoint operator and its continuous form map $\mathcal{V} \rightarrow \mathcal{V}^*$ when the domain makes the meaning unambiguous.

Assumption 3.1 (Common physical form and relative work). On $[0, T]$, $A(t)$ is associated with a closed Hermitian form a_t on the same $\mathcal{V}$. There is a real bounded absolutely continuous scalar $c(t)$ such that

$$k_t = a_t + c(t)\langle \cdot, \cdot \rangle, \quad K(t) = A(t) + c(t) \geq I.$$

The k_t norms are uniformly equivalent to one fixed form norm. For fixed $u, v \in \mathcal{V}$, $k_t(u, v)$ is absolutely continuous, with a strongly measurable form derivative and its integral identity, satisfying

$$|\dot{k}_t(u, v)| \leq \nu(t) \|K(t)^{1/2}u\| \|K(t)^{1/2}v\|, \quad \nu \geq 0, \quad \nu \in L^1(0, T). \quad (3.1)$$

Strong measurability here includes that the derivative applied to any fixed form vector is measurable in $\mathcal{V}^*$.

The shift is an estimate of positivity, not a spectral gap at an incident energy. Nor does this assumption require an operator-norm derivative of a point-Coulomb force. Classical common-form evolution under smoother form hypotheses goes back to Kiszyński [8]. We give the conforming argument for the precise absolutely continuous relative-work assumptions used here.

Lemma 3.2 (Homogeneous common-form propagation). *Under Assumption 3.1 there is a unique unitary propagator $U(t, s)$ on $\mathcal{H}$. It preserves $\mathcal{V}$ and, for $s \leq t$,*

$$\|K(t)^{1/2}U(t, s)v\| \leq \exp\left(\frac{1}{2} \int_s^t \nu(r) dr\right) \|K(s)^{1/2}v\|, \quad v \in \mathcal{V}. \quad (3.2)$$

For $v \in \mathcal{V}$ its trajectory is continuous in $\mathcal{H}$, bounded in $\mathcal{V}$, and solves $i\partial_t U(t, s)v = A(t)U(t, s)v$ in $\mathcal{V}^$.*

Proof. Fix a reference form K_* and let $P_n = 1_{[1, n]}(K_*)$. These need not have finite rank. They are contractions in its form and dual norms, converge strongly there, and their ranges lie in $\mathcal{V}$. On $\mathcal{H}_n = P_n\mathcal{H}$, the restricted forms $K_n(t)$ and $A_n(t) = K_n(t) - c(t)I$ are bounded self-adjoint operators. Uniform equivalence gives their local uniform operator bounds. The integral form identity and (3.1) give absolute continuity in operator norm on each such subspace. The integral equation for the bounded-operator ODE is solved by successive approximations on short intervals and concatenation; its Hermitian generator gives a unitary evolution.

Let $u_n(s) = P_n v$. Differentiation of its form energy yields

$$\frac{d}{dt} \langle u_n, K_n u_n \rangle = \dot{k}_t(u_n, u_n).$$

The two evolution terms cancel because $A_n = K_n - cI$. Applying (3.1) and Gronwall gives (3.2) for u_n , with $P_n v$ at the entrance. In particular u_n is uniformly bounded in $\mathcal{V}$. Its derivative, viewed in the full dual by testing against $P_n w$, is uniformly bounded in $\mathcal{V}^*$.

Weak compactness, followed by a diagonal argument on a countable dense set of form test vectors, gives a limit u weakly continuous in $\mathcal{H}$ and weakly bounded in $\mathcal{V}$, with $u(s) = v$. In the integrated equation each test $P_n w$ converges strongly in $\mathcal{V}$; uniform boundedness of the forms therefore gives $i\dot{u} = A(t)u$ in $\mathcal{V}^*$. At every fixed time weak lower semicontinuity of k_t gives the asserted energy bound. This passage uses conforming compressions of the same forms.

We recall explicitly the norm fact that makes this weak construction unique. If $w \in L^2(\mathcal{V})$ and $\dot{w} \in L^2(\mathcal{V}^*)$, then

$$\|w(t)\|^2 - \|w(s)\|^2 = 2 \operatorname{Re} \int_s^t \langle \dot{w}(r), w(r) \rangle dr, \quad (3.3)$$

and w has a continuous $\mathcal{H}$ representative. One obtains this by time smoothing: for smooth $\mathcal{V}$ -valued functions it is the product rule; Cauchy–Schwarz in the dual pairing passes the integral under convergence in $L^2(\mathcal{V})$ and $W^{1,2}(\mathcal{V}^*)$. The same product rule, integrated from a time whose norm is bounded by its time average, bounds the supremum of the $\mathcal{H}$ norm by these two space norms. It consequently supplies continuous traces and justifies the endpoint passage as well.

Apply (3.3) to the limit and to differences of solutions. Hermiticity gives $\operatorname{Re} \langle -iA(t)w, w \rangle = 0$, proving norm conservation and uniqueness before any strong-convergence claim. Construction backwards from any terminal form vector gives an inverse evolution. Density extends the isometries to mutually inverse unitaries on $\mathcal{H}$. Uniqueness gives their composition rule and strong continuity. The displayed energy bound proves invariance of $\mathcal{V}$. It also gives weak measurability there; separability gives the strong measurability needed for the form-valued integrals below. $\square$

Theorem 3.3 (Ordered inverse dressing for dual forcing). *Under Assumption 3.1, let $q_0 \in \mathcal{V}$ and $f \in W^{1,1}(0, T; \mathcal{V}^*)$. Define*

$$\ell(t) = K(t)^{-1/2} f(t), \quad h(t) = K(t)^{-1/2} \dot{f}(t), \quad N(t) = \int_0^t \nu(s) ds.$$

Then $i\dot{q} = A(t)q + f$, $q(0) = q_0$, has a unique solution in $C([0, T]; \mathcal{H}) \cap L^\infty(0, T; \mathcal{V})$. The inverse lift $g = K^{-1}f$ belongs to $W^{1,1}(0, T; \mathcal{V})$ and obeys the ordered identity

$$\dot{g} = K^{-1} \dot{f} - K^{-1} \dot{K} K^{-1} f, \quad K^{1/2} \dot{g} = h - C_t \ell, \quad C_t = K^{-1/2} \dot{K} K^{-1/2}. \quad (3.4)$$

For $z = q + g$ and $\mathcal{E}_z(t) = \|K(t)^{1/2} z(t)\|$,

$$\mathcal{E}_z(t) \leq e^{N(t)/2} \left[\mathcal{E}_z(0) + \int_0^t e^{-N(s)/2} \{(|c| + \nu)\|\ell\| + \|h\|\}(s) ds \right], \quad (3.5)$$

$$\|K(t)^{1/2} q(t)\| \leq \mathcal{E}_z(t) + \|\ell(t)\|. \quad (3.6)$$

The actual entrance term is $\mathcal{E}_z(0) = \|K(0)^{1/2}(q_0 + K(0)^{-1}f(0))\|$.

Proof. Uniform equivalence and coercivity make each $K(t)$ a bounded isomorphism $\mathcal{V} \rightarrow \mathcal{V}^*$. The exact inverse difference identity is

$$K(t)^{-1} - K(s)^{-1} = -K(t)^{-1}[K(t) - K(s)]K(s)^{-1}.$$

The relative-work bound makes its norm difference bounded by a constant times $\int_s^t \nu$. For a fixed dual vector, the resulting inverse path is absolutely continuous into the Hilbert space $\mathcal{V}$. Strong differentiation of the form identity on fixed vectors, the inverse difference identity, and uniform boundedness therefore give $(K^{-1})'f = -K^{-1}\dot{K}K^{-1}f$ on fixed vectors. For completeness, one can choose a common full-measure set first on a countable dense set of form vectors, then use the integrable relative bound and Lebesgue differentiation to extend the difference quotient to each vector. Approximating the absolutely continuous path f by simple derivatives proves the product rule for $K^{-1}f$. This establishes (3.4) in $\mathcal{V}$. Its second equality is a form-Riesz identification; no derivative of $K^{-1/2}$ has been taken.

Since $A = K - cI$, direct substitution gives

$$i\dot{z} = A(t)z + r, \quad r = cg + i\dot{g} \in L^1(0, T; \mathcal{V}). \quad (3.7)$$

Indeed

$$\|K^{1/2}r\| \leq (|c| + \nu)\|\ell\| + \|h\|.$$

These terms are integrable: f is bounded in the fixed dual norm, $\dot{f}$ is integrable, and the form norms are uniformly equivalent.

Use Lemma 3.2 to define the form-valued Duhamel integral

$$z(t) = U(t, 0)(q_0 + g(0)) - i \int_0^t U(t, s)r(s) ds.$$

Its form norm is bounded by the right side of (3.5). Fubini in the form-dual weak equation verifies (3.7); the usual Hilbert-space Duhamel integral is the same vector, so z is continuous in $\mathcal{H}$. Subtracting g gives the asserted solution q . The difference of any two such solutions has derivative $-iA(t)w$ in $L^\infty(\mathcal{V}^*)$; the norm chain rule makes it zero when its entrance is zero. This proves uniqueness and both inequalities. $\square$

The estimate trades temporal regularity in the form dual for spatial first-form control. It does not require $f \in \mathcal{V}$ or $Af \in \mathcal{H}$. For an actual coupling $f = L(t)p(t)$, sufficient explicit input contracts are

$$\begin{aligned} \|K^{-1/2}LF^{-1}\| &\leq b(t), & \|K^{-1/2}\dot{L}F^{-1}\| &\leq b_1(t), \\ \|\ell\| &\leq b\|Fp\|, & \|h\| &\leq b_1\|Fp\| + b\|F\dot{p}\|. \end{aligned} \quad (3.8)$$

Here $F \geq I$ is a fixed input graph and the products and derivatives must hold on their stated domains. The last quantity is a property of the coupled input, not of a frozen occupation. A changing F adds its own derivative or commutator.

Corollary 3.4 (Retained internal right clock). *For a Hilbert-Schmidt factor satisfying $i\dot{q} = Aq - qD + f$, with fixed bounded Hermitian right clock D , the same argument applies with*

$$r = K^{-1}\{i\dot{f} + cf + fD - i\dot{K}K^{-1}f\}.$$

In (3.5) its inhomogeneous norm can be bounded by $\|K^{-1/2}(i\dot{f} + cf + fD)\|_{\text{HS}} + \nu\|\ell\|_{\text{HS}}$.

Proof. With $z = q + g$, substitution gives $i\dot{z} = Az - zD + cg + gD + i\dot{g}$. Right multiplication by the clock unitary removes $-zD$ and preserves Hilbert-Schmidt norms, including the physical left form norm. Apply the preceding Duhamel estimate and substitute (3.4). $\square$

Only a bounded physical-coordinate-independent internal clock is covered by this corollary. An unbounded right source clock needs its own graph contract. A positional purifier cannot be changed into an internal index just because the Hilbert-space expressions have the same size. The scalar c is not free: it appears in the full combination $i\dot{f} + cf + fD$, while $\dot{c}$ already belongs to $\dot{K}$. Phases may cancel a term only through that complete physical combination.

3.1 Two sharp boundaries of the temporal argument

Example 3.5 (Continuous dual forcing need not give a wave). On $\ell^2(n \geq 2)$ let $A_n = n^2$, $K_n = n^2 + 1$, and $\ell_n(t) = n^{-1}e^{-in^2t}$. Dominated convergence makes $\ell(t)$ continuous in ℓ^2 ; hence $f = K^{1/2}\ell$ is continuous in $\mathcal{V}^*$. Its zero-initial causal solution has components

$$q_n(t) = -it \frac{\sqrt{n^2 + 1}}{n} e^{-in^2t}.$$

For every $t > 0$ these are not square summable. Thus a bounded form-dual coupling and continuous input alone do not give a physical Hilbert-space response. The time-derivative hypothesis in Theorem 3.3 fails.

Proof. The dual group is diagonal, so substitution in the causal integral gives the displayed formula. Its component magnitude tends to t . On the other hand the derivative of ℓ_n has magnitude n , so it cannot supply the required integrable dual derivative of f . $\square$

Proposition 3.6 (Why an inverse square root is not the lift). *The relative form-speed condition does not give a dimension-independent bound on $K^{1/2}(K^{-1/2})'$. It does give an ordered bounded inverse derivative as in (3.4).*

Proof. In a finite eigenbasis, write $\dot{K} = K^{1/2}CK^{1/2}$ and $K_{ii} = \lambda_i > 0$. Differentiate $(K^{1/2})^2 = K$ and then its inverse; the Sylvester equation gives

$$\begin{aligned} (K^{-1/2})'_{ij} &= -\frac{C_{ij}}{\sqrt{\lambda_i} + \sqrt{\lambda_j}}, \\ [K^{1/2}(K^{-1/2})']_{ij} &= -\frac{\sqrt{\lambda_i}C_{ij}}{\sqrt{\lambda_i} + \sqrt{\lambda_j}}. \end{aligned}$$

The first kernel equals $-\int_0^\infty e^{-sK^{1/2}} C e^{-sK^{1/2}} ds$, so its norm is at most $\|C\|/(2\sqrt{\lambda_{\min}})$. The warning concerns the second, weighted derivative.

Take $\lambda_j = 4^j$ and the Hermitian matrix $C_{ij} = i/[\pi(i-j)]$ for $i \neq j$, $C_{ii} = 0$, on indices $1, \dots, N$. Its norm is at most one: it is a compression of the convolution operator on $\ell^2(\mathbb{Z})$ whose Fourier multiplier is, up to its sign, $(\theta - \pi)/\pi$ on $(0, 2\pi)$. This follows by integrating the Fourier series of the sawtooth, or by Abel summing $\sum_{d \neq 0} e^{id\theta}/d = i(\pi - \theta)$. For the unit vector $v = N^{-1/2}(1, \dots, 1)$, pairing the two off-diagonal entries at separation d gives

$$\left| \langle v, K^{1/2}(K^{-1/2})'v \rangle \right| = \frac{1}{\pi N} \sum_{d=1}^{N-1} \frac{(N-d) \tanh(d \log 2/2)}{d}.$$

For $2 \leq d \leq N/2$, the hyperbolic tangent is at least $3/5$ and $(N-d)/N \geq 1/2$. The harmonic sum diverges, proving the assertion. These are actual positive paths $K(s) = K(0)^{1/2}(I + sC)K(0)^{1/2}$, $|s| \leq 1/2$. Since $K(0) \geq 4I$, these obey $K(s) \geq 2I$; their relative speed is bounded by $(1 - |s|)^{-1} \leq 2$. The ordered full inverse in the theorem avoids this weighted-square-root obstruction. $\square$

4 A finite spectral window with a varying coherent input

We next fix the complete self-adjoint complement A . Its Hilbert space may contain continua and all quantum source coordinates. Let $\mathcal{U}$ be the input Hilbert space, $B : \mathcal{U} \rightarrow \mathcal{H}$ bounded, and $a \in L^2(0, T; \mathcal{U})$. The zero-initial response is

$$q_B(t) = -i \int_0^t e^{-iA(t-s)} B a(s) ds. \quad (4.1)$$

A spectral sector can be included by replacing B with CB for a spectral projection C of A . The input may change direction arbitrarily in $\mathcal{U}$.

Lemma 4.1 (An operator spectral measure on one interval). *Let $I = [\alpha, \alpha + l)$, $l > 0$, and suppose $B^*E_A(I)B \leq mI_{\mathcal{U}}$. For $v \in H^1(I; \mathcal{U})$, the spectral integral $\mathcal{T}_I v = \int_I E_A(d\lambda)Bv(\lambda)$ satisfies*

$$\|\mathcal{T}_I v\|^2 \leq 2m \left\{ l^{-1} \|v\|_{L^2(I)}^2 + l \|v'\|_{L^2(I)}^2 \right\}. \quad (4.2)$$

Proof. Let $\bar{v} = l^{-1} \int_I v$ be the Bochner mean. The fundamental theorem for Hilbert-valued H^1 functions gives

$$v(\lambda) = \bar{v} + \int_I k(\lambda, \xi) v'(\xi) d\xi, \quad k(\lambda, \xi) = 1_{\xi < \lambda} - \frac{\alpha + l - \xi}{l}, \quad |k| \leq 1.$$

For every scalar b supported in I with $|b| \leq 1$,

$$\|b(A)E_A(I)B\|^2 = \|B^*E_A(I)|b(A)|^2B\| \leq m.$$

Consequently the spectral integral has the precise order

$$\mathcal{T}_I v = E_A(I)B\bar{v} + \int_I k(A, \xi)E_A(I)Bv'(\xi) d\xi.$$

For smooth v this follows by its displayed integral representation and bounded spectral calculus. The same formula defines a continuous extension to H^1 . It gives

$$\|\mathcal{T}_I v\| \leq \sqrt{m} \left\{ l^{-1/2} \|v\|_{L^2(I)} + l^{1/2} \|v'\|_{L^2(I)} \right\}.$$

Squaring and using $(x + y)^2 \leq 2x^2 + 2y^2$ proves (4.2). No common eigenvector of the input-valued function or simultaneous diagonalization of the operator measure was assumed. $\square$

Theorem 4.2 (Finite-imaginary-part response window). *For $\eta > 0$ define the positive operator*

$$M_\eta(E) = \frac{1}{\pi} B^* \frac{\eta}{(A - E)^2 + \eta^2} B. \quad (4.3)$$

Suppose $M_\eta(E) \leq s_\eta I$ for all real E , or at the centers of a partition into intervals of length 2η covering the relevant spectrum. Then, for $0 \leq t \leq T$,

$$\|q_B(t)\|^2 \leq 4\pi^2 s_\eta (1 + \eta^2 t^2) \int_0^t \|a(s)\|^2 ds. \quad (4.4)$$

Atoms, singular spectral measures and thresholds are allowed. The coefficient is an operator ceiling for the complete input space, not a scalar response for one incident column.

Proof. On $I_j = [E_j - \eta, E_j + \eta)$ the Poisson kernel in (4.3) is at least $1/(2\pi\eta)$. Hence $B^*E_A(I_j)B \leq 2\pi\eta s_\eta I$. Set

$$v_t(\lambda) = \int_0^t e^{i\lambda(s-t/2)} a(s) ds.$$

This Bochner Fourier transform is in $H^1(\mathbb{R}; \mathcal{U})$, even though no derivative of a was required. Hilbert-valued Plancherel gives

$$\begin{aligned} \|v_t\|_{L^2(\mathbb{R})}^2 &= 2\pi \int_0^t \|a(s)\|^2 ds, \\ \|v_t'\|_{L^2(\mathbb{R})}^2 &= 2\pi \int_0^t (s - t/2)^2 \|a(s)\|^2 ds \leq \frac{t^2}{4} \|v_t\|_{L^2(\mathbb{R})}^2. \end{aligned}$$

The full spectral integral of v_t differs from $q_B(t)$ only by $-ie^{-iAt/2}$. This identity follows directly by Fubini from (4.1). The outputs $\mathcal{T}_{I_j}v_t$ lie in mutually orthogonal spectral subspaces. Apply Lemma 4.1 with $l = 2\eta$, $m = 2\pi\eta s_\eta$ and sum:

$$\|q_B(t)\|^2 \leq 2\pi s_\eta \|v_t\|_2^2 + 8\pi\eta^2 s_\eta \|v'_t\|_2^2.$$

The Plancherel identities give (4.4). Orthogonality and monotone summation justify infinitely many intervals without a dimension or number-of-channels factor. $\square$

The interval derivative term is substantive. A ceiling on $B^*E_A(I)B$ by itself cannot bound a spectral integral of a changing vector by a dimension-free multiple of $\sup_\lambda \|v(\lambda)\|^2$. The time-centered Fourier representation is what supplies the derivative in this theorem. It is a derivative in spectral energy, not a time-regularity premise on the actual input.

For intervals of half-width δ instead of η , the same proof gives

$$\|q_B(t)\|^2 \leq 2\pi^2 s_\eta \frac{\eta^2 + \delta^2}{\eta} (\delta^{-1} + \delta t^2) \int_0^t \|a(s)\|^2 ds. \quad (4.5)$$

Indeed the interval measure costs $m = \pi(\eta^2 + \delta^2)s_\eta/\eta$ and its length is 2δ . Thus the observation window can be chosen to match an earned response table.

If $B = LF^{-1}$, use $a = Fp$ with the actual input. For $F = I$ and a normalized full block evolution, $\int_0^t \|p(s)\|^2 ds \leq t$; choosing $\eta = 1/T$ gives the uniform ceiling $8\pi^2 s_{1/T} T$. If F is dimensionless and L, A have inverse-time units, s_η has inverse-time units, so this norm bound is dimensionless. An atom of response weight w at λ_0 has $M_\eta(\lambda_0) = w/(\pi\eta)$, whereas a resonant constant input can give response wt^2 . The theorem retains that pole through s_η ; it does not convert it into a decay rate or a gap.

4.1 Unbounded form coupling: near response and far dressing

Theorem 4.3 (Coherent near/far response in the physical form). *Let A be fixed self-adjoint, $K = A + c \geq I$, and $F \geq I$ a fixed positive input graph. Assume*

$$B = K^{-1/2}LF^{-1} : \mathcal{U} \longrightarrow \mathcal{H} \quad \text{is bounded.}$$

Thus $LF^{-1} = K^{1/2}B$ is a bounded map into $\mathcal{V}^$, possibly not into $\mathcal{H}$. Assume $Fp \in L^2(0, T; \mathcal{U})$ for the near-response statement. For $J_0 = [E_0 - \Lambda, E_0 + \Lambda]$, $\Lambda > 0$, let $C = 1_{J_0}(A)$. For $j = 0, 1$ set*

$$B_j = K^{(j+1)/2}CB, \quad M_{j,\eta}(E) = \frac{1}{\pi} B^* C K^{j+1} \frac{\eta}{(A - E)^2 + \eta^2} CB.$$

These B_j are bounded and $M_{j,\eta}$ are positive operator spectral responses. If $M_{j,\eta}(E) \leq s_{j,\eta}I$ at the required centers, the zero-initial near response satisfies

$$\|K^{j/2}q_{\text{near}}(t)\|^2 \leq 4\pi^2 s_{j,\eta} (1 + \eta^2 t^2) \int_0^t \|Fp(s)\|^2 ds. \quad (4.6)$$

Let

$$R_j = K^{(j+1)/2}(A - E_0)^{-1}(1 - C)B.$$

If $a_0(s) = e^{iE_0 s}Fp(s) \in W^{1,1}(0, t; \mathcal{U})$, the far response satisfies

$$\|K^{j/2}q_{\text{far}}(t)\| \leq \|R_j\| \left\{ \|a_0(t)\| + \|a_0(0)\| + \int_0^t \|\dot{a}_0(s)\| ds \right\}, \quad (4.7)$$

$$\|R_0\| \leq \frac{\sqrt{\Lambda + |E_0 + c|}}{\Lambda} \|B\|, \quad \|R_1\| \leq \left(1 + \frac{|E_0 + c|}{\Lambda}\right) \|B\|. \quad (4.8)$$

An original $q_0 \in \mathcal{V}$ contributes the separate homogeneous term $e^{-iAt}q_0$, with its original form norm. It is not part of (4.6).

Proof. The causal integral is first defined in $\mathcal{V}^*$, where e^{-iAt} is a unitary group in the K dual norm. On the bounded spectral interval, multiplication by $K^{j/2}C$ turns its forcing into $B_j Fp$. Apply Theorem 4.2 to this bounded coupling to obtain (4.6).

On the far sector put $D = A - E_0$. Commuting only functions of this same self-adjoint A , integration by parts gives the exact identity

$$K^{j/2}q_{\text{far}}(t) = e^{-iE_0 t} \left\{ e^{-iDt} R_j a_0(0) - R_j a_0(t) + \int_0^t e^{-iD(t-s)} R_j \dot{a}_0(s) ds \right\}. \quad (4.9)$$

For an unbounded coupling, prove it first on bounded spectral cutoffs. The bounds below pass it to the form-dual causal solution and also show that its right side is a physical $K^{j/2}$ vector. Taking norms proves (4.7). The two displayed endpoints are the virtual dressings; neither can be dropped.

For $x = |\lambda - E_0| \geq \Lambda$ and $\lambda \in \sigma(A)$, positivity and the triangle inequality give $0 < \lambda + c \leq x + |E_0 + c|$. Therefore

$$\frac{\sqrt{\lambda + c}}{|\lambda - E_0|} \leq \frac{\sqrt{\Lambda + |E_0 + c|}}{\Lambda}, \quad \frac{\lambda + c}{|\lambda - E_0|} \leq 1 + \frac{|E_0 + c|}{\Lambda}.$$

The spectral theorem proves (4.8) and the limiting integration-by-parts assertion. The homogeneous term follows by linearity; it has unchanged K norm because $K = A + c$ commutes with A . $\square$

One may use $M_{1,\eta} \leq k_{J_0} M_{0,\eta}$, where $k_{J_0} = \sup_{\lambda \in J_0 \cap \sigma(A)} (\lambda + c)$, if that additional factor is charged. A smaller first-form response coefficient is a separate spectral calculation. For several coherent couplings $L(t) = \sum_{\alpha} l_{\alpha}(t) L_{\alpha}$, apply the theorem to the stacked bounded map with input $(l_1 Fp, \dots, l_m Fp)$. Its full operator measure retains all off-diagonal responses. Summing independent scalar channel rates would be a different statement.

The near estimate requires an L^2 input but no input time derivative. The far estimate requires the displayed derivative and endpoint dressings. Neither frozen spectral statement applies automatically to a changing $A(t)$. One must instead use Theorem 3.3, or prove a comparison with a fixed parent and charge its actual forcing and form-dual derivative.

5 Ordered resolvents for an unbounded closed-form interaction

Theorem 5.1 (Closed-form factor transfer and physical weights). *Let A_0 and $A = A_0 + W^* J W$ be self-adjoint operators defined by closed semibounded forms on the same $\mathcal{V}$, with equivalent positive shifted form norms. Here $W : \mathcal{V} \rightarrow \mathcal{H}_{\text{aux}}$ is bounded and $J = J^*$ is bounded on $\mathcal{H}_{\text{aux}}$. The adjoint W^* maps $\mathcal{H}_{\text{aux}}$ to $\mathcal{V}^*$. Actual form closure is an assumption; a formal factorization alone does not supply it.*

For $\zeta = E + i\eta$, $\eta > 0$, set

$$R_0 = (A_0 - \zeta)^{-1}, \quad R = (A - \zeta)^{-1}, \quad G_0 = W R_0 W^*, \quad G = W R W^*.$$

All products are bounded between the indicated form spaces, and

$$T = (I + J G_0)^{-1} = I - J G, \quad R W^* = R_0 W^* T, \quad G = G_0 T. \quad (5.1)$$

In particular

$$\operatorname{Im} G = T^*(\operatorname{Im} G_0)T = \eta(RW^*)(RW^*), \quad (5.2)$$

$$(RW^*)^*T_{\text{ph}}(RW^*) = T^*[(R_0W^*)^*T_{\text{ph}}(R_0W^*)]T. \quad (5.3)$$

The second formula is an equality of bounded auxiliary-space forms for any positive closed physical form T_{ph} with $\mathcal{V} \subset \operatorname{Dom} T_{\text{ph}}^{1/2}$ continuously. No commutation with A_0, A, W or J is required.

Proof. Choose $K_0 = A_0 + c_0 \geq I$. The spectral theorem gives the exact identity

$$\|K_0^{1/2}R_0K_0^{1/2}\| = \sup_{\lambda \in \sigma(A_0)} \frac{\lambda + c_0}{|\lambda - \zeta|} < \infty. \quad (5.4)$$

Thus R_0 extends boundedly $\mathcal{V}^* \rightarrow \mathcal{V}$. The same argument with a positive shift of A and equivalence of form norms gives this extension for R . Their form inverse identities therefore give both ordered resolvent identities

$$R - R_0 = -R_0W^*JWR = -RW^*JWR_0.$$

Sandwiching by W, W^* yields $G - G_0 = -G_0JG = -GJG_0$. It follows by multiplication in the displayed order that both products of $I + JG_0$ and $I - JG$ equal I . This proves invertibility and the formula for T . The first resolvent identity now gives $RW^* = R_0W^*(I - JG)$, proving (5.1).

For a self-adjoint resolvent, $\operatorname{Im} R = \eta R^*R$. The identity extends to dual inputs by continuity: both sides define bounded sesquilinear forms on $\mathcal{V}^*$, and Hilbert inputs are dense there. Sandwiching with W^* proves the second equality in (5.2). Substitution of $RW^* = R_0W^*T$ proves its first equality. The same substitution into the quadratic form of T_{ph} gives (5.3). Continuity of that form on $\mathcal{V}$ justifies each product even when T_{ph} is unbounded on $\mathcal{H}$. $\square$

The resolvent inverse in this theorem exists off the real axis because the closed self-adjoint parents exist. A Neumann criterion is sufficient for a quantitative bound, not necessary for the identity:

$$\|JG_0\| \leq \theta < 1 \implies \|T\| \leq (1 - \theta)^{-1}. \quad (5.5)$$

If $LF^{-1} = W^*b$ with bounded b , an earned $\operatorname{Im} G_0/\pi \leq s_{0,\eta}I$ implies

$$\frac{1}{\pi} \operatorname{Im}[(LF^{-1})^*R(LF^{-1})] \leq \frac{\|b\|^2 s_{0,\eta}}{(1 - \theta)^2} I$$

when its full form-dual sandwich is used. Likewise (5.3) transfers an earned physical-energy resolvent bound. Absorptive response alone does not replace that physical-energy bound. For a form-dual coupling this is first a resolvent sandwich: one uses the bounded near factors of Theorem 4.3, or the ordered temporal estimate below, rather than treating LF^{-1} as a bounded Hilbert-space coupling without proof.

Equality in (5.4) is over the actual spectrum. Replacing it by a whole half-line above a spectral floor gives an upper bound, not in general an equality. Spectral gaps matter for that distinction, whereas the shift that makes K_0 positive does not create such gaps. At a real eigenvalue no inverse has been asserted. As η decreases, genuine poles and thresholds may make both the baseline response and the quantitative inverse ceiling large.

6 Damped temporal response with a noncommuting physical form

The physical derivative form generally does not commute with the interacting complement. It may therefore be placed outside an actual resolvent, as below, but cannot be inserted into a scalar spectral measure as though it commuted with every spectral projection. This use of a resolvent and a time integral is related to the Kato smoothing framework [6, 7]. The statement here is at one earned $\eta > 0$; it does not assert the uniform imaginary-part bounds required by a global smoothing theorem.

Theorem 6.1 (Ordered damped response). *Let A be fixed self-adjoint, $K = A + c \geq I$, and $\mathcal{L} = LF^{-1} : \mathcal{U} \rightarrow \mathcal{V}^*$ bounded. Let $T_{\text{ph}} \geq I$ be a positive closed form whose domain contains $\mathcal{V}$ continuously. For fixed $\eta > 0$ suppose the actual ordered sandwich satisfies the measurable bound*

$$\|T_{\text{ph}}^{1/2}(A - E - i\eta)^{-1}\mathcal{L}\|^2 \leq S_\eta(E). \quad (6.1)$$

For $a \in L^2(0, T; \mathcal{U})$, extend the forcing by zero after T , and let q be its causal zero-initial form-dual response. Define $a_\eta(s) = e^{-\eta s}a(s)1_{[0, T]}(s)$ and use the unitary Fourier convention $\hat{u}(E) = (2\pi)^{-1/2} \int_{\mathbb{R}} e^{iEt}u(t) dt$. If the right side is finite, then

$$\int_0^\infty e^{-2\eta t} \|T_{\text{ph}}^{1/2}q(t)\|^2 dt \leq \int_{\mathbb{R}} S_\eta(E) \|\hat{a}_\eta(E)\|^2 dE. \quad (6.2)$$

In particular a uniform $S_\eta(E) \leq S_\eta$ gives the upper bound $S_\eta \|a_\eta\|_2^2$. A bound $S_\eta(E) \leq S_0 + S_1 E^2$ instead gives

$$S_0 \|a_\eta\|_2^2 + S_1 \|\dot{a}_\eta\|_2^2, \quad (6.3)$$

provided the actual zero extension a_η belongs to $H^1(\mathbb{R}; \mathcal{U})$.

Proof. The fixed group e^{-iAt} is unitary also on the K form dual. The causal integral

$$q(t) = -i \int_0^{\min(t, T)} e^{-iA(t-s)} \mathcal{L}a(s) ds$$

is therefore a continuous $\mathcal{V}^*$ vector. It is zero for negative times and bounded in that dual norm for all positive times. Its damped extension $q_\eta = e^{-\eta t}q 1_{[0, \infty)}$ belongs to $L^2(\mathbb{R}; \mathcal{V}^*)$. Its value at zero is zero, so differentiation creates no entrance delta distribution. Fourier transformation of the causal equation, or Fubini applied to its damped convolution, gives the exact form-dual identity

$$\hat{q}_\eta(E) = -(A - E - i\eta)^{-1} \mathcal{L} \hat{a}_\eta(E). \quad (6.4)$$

For almost every E the right side lies in $\mathcal{V}$, hence in $\text{Dom } T_{\text{ph}}^{1/2}$. The assumed finite integral and (6.1) place it in the L^2 graph of $T_{\text{ph}}^{1/2}$. Coercivity $T_{\text{ph}} \geq I$ also places it in physical $L^2(\mathcal{H})$. Fourier transformation commutes with this closed spatial operator: to see this, replace it by its bounded spectral truncations, use Hilbert-valued Plancherel there, and pass by its closed graph and monotone convergence of the graph norms. Thus inverse Fourier transformation identifies this same dual solution as a physical $L^2(\text{Dom } T_{\text{ph}}^{1/2})$ response. Plancherel and (6.1) prove (6.2). In this argument the truncations are of T_{ph} for the Fourier graph identity; none is commuted through A or its resolvent.

For a uniform coefficient use Plancherel on a_η . For the quadratic coefficient use $\|E \hat{a}_\eta\|_2 = \|\dot{a}_\eta\|_2$. That equality requires the stated whole-line H^1 condition and proves (6.3). $\square$

The theorem gives an integrated first-form response, not an all-times pointwise form bound. If the input is H^1 inside $[0, T]$ but has nonzero endpoint values, its zero extension has delta derivatives and is not $H^1(\mathbb{R})$. One must keep the endpoint dressings, as in (4.9) or Theorem 3.3, or prove a physical pulse collar that removes those jumps. The endpoint problem cannot be repaired by simply omitting the positive frequency-weighted term.

The physical finite-interval graph norm obeys

$$\|T_{\text{ph}}^{1/2} q\|_{L^2(0,T)} \leq e^{\eta T} \left(\int_{\mathbb{R}} S_{\eta}(E) \|\widehat{a}_{\eta}(E)\|^2 dE \right)^{1/2}. \quad (6.5)$$

If $q_0 \neq 0$, its homogeneous response must be added separately. For example, an earned $T_{\text{ph}} \leq \kappa K$ and $q_0 \in \mathcal{V}$ give

$$\left(\int_0^\infty e^{-2\eta t} \|T_{\text{ph}}^{1/2} e^{-iAt} q_0\|^2 dt \right)^{1/2} \leq \sqrt{\frac{\kappa}{2\eta}} \|K^{1/2} q_0\|.$$

Add this to the square root of the forced bound; the square of the sum, not the sum of squares, is a generally valid full-response bound. For a noncoercive derivative form, replace it by $I + T_{\text{ph}}$ or establish a separate Hilbert-norm sandwich. Temporal damping has neither projected out a bound pole nor supplied a missing source graph.

7 Whole-form spatial and source derivatives

Temporal smoothing and spatial differentiation make different demands. The preceding theorems do not require a spatial derivative of the forcing when their temporal or resolvent assumptions are earned. A physical source derivative, however, must act on the complete coupled form and the actual input; replacing that coordinate by a classical label changes the problem.

Proposition 7.1 (Ordered whole-form derivative). *Let A and $\mathcal{L} = LF^{-1}$ be as in Theorem 6.1. Let δ_i be either a parameter derivative on the common form domain, or a closed physical derivation $\delta_i M = D_i^Q M - M D_i^P$. For a parameter family, require strong difference-quotient convergence of A in $\mathcal{L}(\mathcal{V}, \mathcal{V}^*)$ and of $\mathcal{L}$ in $\mathcal{L}(\mathcal{U}, \mathcal{V}^*)$, with locally uniform bounds on these quotients and uniformly equivalent nearby form norms. For a physical derivation, require domain-preserving translations or local gauges for which the conjugated families satisfy these same conditions and realize the displayed derivatives on a common core. Suppose the actual whole-form sandwiches are bounded:*

$$\|K^{-1/2}(\delta_i A)K^{-1/2}\| \leq \alpha_i, \quad \|K^{-1/2}\delta_i \mathcal{L}\| \leq \beta_i. \quad (7.1)$$

For $\zeta = E + i\eta$, $\eta > 0$, write $R_\zeta = (A - \zeta)^{-1}$ and $U_\zeta = K^{1/2}R_\zeta \mathcal{L}$. Then

$$\delta_i(R_\zeta \mathcal{L}) = R_\zeta \{\delta_i \mathcal{L} - (\delta_i A)R_\zeta \mathcal{L}\}, \quad (7.2)$$

$$\|K^{1/2}\delta_i(R_\zeta \mathcal{L})\| \leq J_\zeta \{\beta_i + \alpha_i \|U_\zeta\|\}, \quad J_\zeta = \sup_{\lambda \in \sigma(A)} \frac{\lambda + c}{|\lambda - \zeta|}. \quad (7.3)$$

If ζ itself varies, add $(\delta_i \zeta)R_\zeta^2 \mathcal{L}$ to (7.2).

Proof. For a parameter family, the form resolvent difference identity is

$$R_{\zeta, \epsilon} - R_\zeta = -R_{\zeta, \epsilon}(A_\epsilon - A)R_\zeta.$$

It is a bounded identity from $\mathcal{V}^*$ to $\mathcal{V}$. Strong differentiation on fixed vectors, uniform local form bounds and the given difference-quotient hypothesis yield $\delta_i R_\zeta = -R_\zeta(\delta_i A)R_\zeta$. Leibniz gives (7.2).

For a physical derivation, conjugate by the domain-preserving translations/gauges first and apply the same argument; differentiation on the common core is the commutator formula. Strong form convergence and closedness of the physical derivative pass it to its admitted domain.

Insert a form Riesz map without changing order:

$$K^{1/2}R_\zeta(\delta_i A)R_\zeta\mathcal{L} = [K^{1/2}R_\zeta K^{1/2}][K^{-1/2}(\delta_i A)K^{-1/2}]U_\zeta.$$

The first factor has norm J_ζ by the spectral theorem. The analogous factorization of the $\delta_i\mathcal{L}$ term proves (7.3). Differentiating $A - \zeta$ when ζ varies gives the stated extra term with its positive sign. $\square$

For a literal input vector a , the corresponding physical derivative remains

$$D_i^Q(R_\zeta\mathcal{L}a) = [\delta_i(R_\zeta\mathcal{L})]a + R_\zeta\mathcal{L}D_i^P a. \quad (7.4)$$

Its second term cannot be erased by a response norm estimate. The graph and derivative domain of a must be included. When F is covariantly constant, $\delta_i\mathcal{L} = (\delta_i L)F^{-1}$; otherwise

$$\delta_i\mathcal{L} = (\delta_i L)F^{-1} - LF^{-1}(\delta_i F)F^{-1}$$

on the admitted graph, with an additional bound for that second term. Gauge curvature, moving frames, source kinetic terms and mobility commutators all belong to the whole forms in (7.1). Differentiating a singular factor W is unnecessary and may be invalid even when the derivative of the whole form W^*JW is bounded.

For example, in three relative dimensions the Hardy inequality gives the legitimate whole Coulomb force form

$$|\langle u, \partial_{R_i}(g/|r - R|)v \rangle| \leq |g| \|u/|r - R|\| \|v/|r - R|\| \leq 4|g| \|\nabla_r u\| \|\nabla_r v\|.$$

With an earned $K \geq \lambda_{\text{rel}}(-\Delta_r)$, its relative coefficient is at most $4|g|/\lambda_{\text{rel}}$. The coefficient is the true relative-mass kinetic coefficient; source-dependent center maps add their actual derivatives. This is a form estimate, not an L^2 operator bound on the force. Strong vector continuity or the stated difference-quotient contract suffices; operator-norm continuity under translations is not assumed.

All statements above keep a fixed physical block chart. For a varying projector one must include its domain-preserving transport and the moving frame term $-iU^*\dot{U}$. A nonlocal Hilbert-space band unitary is not automatically an isometry of pointwise physical currents. Likewise a moving hard-core boundary requires a proved domain transport before it enters a fixed-domain force identity.

8 Spatial force and temporal regularity as distinct current resources

The form-dual estimates above and the following estimates solve different input problems. A residual with a spatial form norm can be propagated using actual forces without paying an irrelevant constant carrier energy. A residual with only a Hilbert norm can instead use temporal regularity and a common strong domain. Neither hypothesis implies the other for singular interactions.

8.1 A force estimate with covariant drift

Theorem 8.1 (Forced positive form with complete first-order work). *Use (2.1) and $K = -\sum_i \lambda_i D_i^2 + U \geq I$, with scalar $U \geq 1$. Assume a common form domain, the true unitary propagator, and a form-stable approximation justifying the commutators below. Suppose $K' \leq a_K K$ as forms and, for form vectors v ,*

$$\begin{aligned} \left(\sum_i \|\sqrt{\lambda_i}(\nabla_i V)v\|^2 \right)^{1/2} &\leq g_0 \|v\| + g_1 \|K^{1/2}v\|, \\ \|B\| &\leq L, \quad B_{ji} = \sqrt{\lambda_j/\lambda_i} \nabla_j a_i, \\ \left(\sum_j \|\sqrt{\lambda_j} A_j v\|^2 \right)^{1/2} &\leq G_0 \|v\| + G_1 \|K^{1/2}v\|, \\ \left\| U^{-1/2} \sum_i a_i (\partial_i U) v \right\| &\leq u_0 \|v\| + u_1 \|K^{1/2}v\|, \end{aligned}$$

where B acts on the coordinate direct sum, and

$$[D_j, D_i] = -i\Omega_{ji}, \quad A_j = \sum_i \left(-ia_i \Omega_{ji} + \frac{1}{2} \nabla_j \nabla_i a_i \right).$$

All scalar coefficients are nonnegative and integrable. If $iE' = HE + r$, $E(0)$ is in the form domain and $r \in L^1([0, T]; \text{Dom } K^{1/2})$, then

$$\begin{aligned} e(t) &\leq e_0 + \int_0^t \|r(s)\| ds =: E_*(t), \\ Q(t) &\leq e^{B_*(t)} \left[Q_0 + \int_0^t e^{-B_*(s)} \left(\beta(s) E_*(s) + \|K(s)^{1/2} r(s)\| \right) ds \right], \\ B_*(t) &= \int_0^t \alpha(s) ds, \quad \alpha = \frac{1}{2} a_K + g_1 + L + G_1 + \frac{1}{2} u_1, \quad \beta = g_0 + G_0 + \frac{1}{2} u_0. \end{aligned} \tag{8.1}$$

Here $e = \|E\|$ and $Q = \|K^{1/2}E\|$.

Proof. On a common smooth core, the product rule gives

$$i\langle v, [V, K]v \rangle = 2 \text{Im} \sum_j \langle \sqrt{\lambda_j} D_j v, \sqrt{\lambda_j} (\nabla_j V) v \rangle.$$

Let $F = -i \sum_i (a_i D_i + \nabla_i a_i / 2)$. In its written matrix order,

$$[D_j, F] = -i \sum_i \left((\nabla_j a_i) D_i + a_i [D_j, D_i] + \frac{1}{2} \nabla_j \nabla_i a_i \right).$$

In the derivative of the kinetic form, the self-adjoint F acting on $D_j E$ cancels, leaving

$$-2 \text{Re} \sum_{ji} \lambda_j \langle D_j E, (\nabla_j a_i) D_i E \rangle - 2 \text{Re} \sum_j \lambda_j \langle D_j E, A_j E \rangle.$$

Their absolute values are at most $2LQ^2$ and $2Q(G_0e + G_1Q)$. Also $i[F, U] = \sum_i a_i \partial_i U$, whose quadratic form is bounded by $Q(u_0e + u_1Q)$. Thus, retaining the forcing,

$$(Q^2)' \leq 2\alpha Q^2 + 2\beta eQ + 2Q\|K^{1/2}r\|.$$

Regularize Q at zero and use Duhamel's norm bound for e . The integrating factor proves (8.1).

For form data, set $R_\epsilon = (I + \epsilon K)^{-1}$ and $K_\epsilon = K(I + \epsilon K)^{-1}$. The ordered identities

$$K'_\epsilon = R_\epsilon K' R_\epsilon, \quad [V + F, K_\epsilon] = R_\epsilon [V + F, K] R_\epsilon$$

give the same inequality with $Q_\epsilon = \|K_\epsilon^{1/2} E\|$. Indeed $\|R_\epsilon E\| \leq e$, $\|K_\epsilon^{1/2} R_\epsilon E\| \leq Q_\epsilon$, and the forcing is bounded by $Q_\epsilon \|K_\epsilon^{1/2} r\|$. Integrate before taking the monotone spectral limit $\epsilon \downarrow 0$. The assumed conforming approximation supplies the propagator and commutator passage for unbounded coefficients. Resolvent algebra alone does not prove those domain premises. $\square$

When $a_i = 0$, this reduces to the spatial-force estimate with $\alpha = a_K/2 + g_1$ and $\beta = g_0$. A covariantly constant carrier $M(t)$ has $\nabla_i M = 0$ and contributes no spatial force. Its phase is still in the true evolution. A laboratory-constant matrix can instead have $[C_i, M] \neq 0$ and must then be priced. For a scalar dilation $a = cx$ in one flat coordinate, the kinetic contribution is $-2c\|\partial_x E\|^2$, confirming the commutator sign. Spatially varying mobility or a discontinuous coefficient needs additional terms or a different theorem.

8.2 A positive invariant when a quadratic trap is inverted

Proposition 8.2 (Transported quadratic form and physical coflow). *Let $H_0(t)$ be a real scalar Weyl-ordered quadratic rate Hamiltonian in $z = (q, p)$, $p = -i\partial_q$, including its affine terms. Let $S(t)$ be its classical symplectic propagator and $m(t)$ its affine classical solution. For any positive matrix W_0 , set*

$$W(t) = S(t)W_0S(t)^T, \quad K(t) = 1 + \frac{1}{4}(z - m)^T W(t)^{-1}(z - m).$$

Then $K \geq 1$ and, as forms on the transported Schwartz core,

$$K_t + i[H_0, K] = 0. \tag{8.2}$$

For a pure Gaussian with phase gradient $m_p + R(q - m_q)$, $R = R^T$, and positive imaginary phase matrix I , its covariance has blocks

$$W = \begin{pmatrix} C & CR \\ RC & RCR + I/2 \end{pmatrix}, \quad C = (2I)^{-1}.$$

The invariant is therefore

$$K = 1 + \frac{1}{2} \left[(p - m_p - R(q - m_q))^T I^{-1} (p - m_p - R(q - m_q)) + (q - m_q)^T I (q - m_q) \right].$$

Writing $D_0 = \partial - i(m_p + R(q - m_q))$ and $\Lambda = \frac{1}{2}I^{-1}$, one has $K \geq D_0^\dagger \Lambda D_0$. The Gaussian ground value of K is $1 + d/2$.

Proof. If the classical quadratic generator is $A = JG_0$, then $W' = AW + WA^T$ and $(W^{-1})' = -A^T W^{-1} - W^{-1}A$. The affine equation for m cancels the linear terms. Commutators of Weyl quadratics equal their Poisson expressions, since higher Moyal derivatives vanish; these identities prove (8.2). A positive quadratic matrix is a sum of squares of real linear canonical observables, proving positivity. Factoring the displayed covariance as

$$\begin{pmatrix} 1 & 0 \\ R & 1 \end{pmatrix} \begin{pmatrix} C & 0 \\ 0 & I/2 \end{pmatrix} \begin{pmatrix} 1 & R \\ 0 & 1 \end{pmatrix}$$

and inverting proves the form formula. Each centered Gaussian coordinate contributes $1/2$ to its quadratic ground value. $\square$

This is established invariant mathematics [12, 13]; its role here is a positive error metric through an inverted interval. It is dimensionless and does not replace the physical mass in the current. For the same laboratory vector potential A_0 , define $b_{0,i} = (\hbar m_{p,i} + \hbar(R(q - m_q))_i - e_i A_{0,i})/M_i$. For any true wave in that constitution,

$$j_i[\Psi] - \rho_\Psi b_{0,i} = \frac{\hbar}{M_i} \text{Im}(\Psi^\dagger D_{0,i} \Psi).$$

Consequently its difference from the analogous expression for P is at most

$$\frac{\hbar}{M_i} \left(e \|D_{0,i} P\| + N \sqrt{(\Lambda^{-1})_{ii}} Q \right) \quad (8.3)$$

in integrated absolute value. The weighted coordinate Cauchy inequality proves the last factor. A changed potential ΔA adds $-e_i \Psi^\dagger \Delta A_i \Psi / M_i$ to the relative current.

For $H = H_0 + V$ with Hermitian multiplication V and form forcing r , the exact cancellation (8.2) gives $Q' \leq g_1 Q + g_0 e + \|K^{1/2} r\|$ if $\|\Lambda^{1/2} \nabla V v\| \leq g_0 \|v\| + g_1 \|K^{1/2} v\|$. The proof is the preceding commutator calculation, with no separate K_t loss. Resolvent regularization preserves $K_{\epsilon,t} + i[H_0, K_\epsilon] = 0$. A canonical drift rewritten in D_0 also creates the multiplication term $a \cdot (m_p + R(q - m_q))$; its force and literal current remain. An internal-branch-dependent covariance would produce further commutators and is outside this scalar invariant statement.

8.3 Temporal transfer with a retained internal clock

Theorem 8.3 (First-domain causal estimate with a form-controlled time derivative). *Let $H(t)$ be self-adjoint on a fixed domain $\mathcal{D}$, with $t \mapsto H(t) \in \mathcal{L}(\mathcal{D}, \mathcal{H})$ continuously differentiable for an equivalent fixed graph norm. Assume its unitary common-domain propagator is locally bounded in that graph norm. Let D_s be a fixed finite Hermitian right-clock matrix, and write $\mathcal{L}(t)X = H(t)X - XD_s$. For an absolutely continuous Hilbert-Schmidt residual r , with $r' \in L^1$, consider*

$$i\delta' = \mathcal{L}\delta - r, \quad \delta(0) = 0.$$

Suppose $K = H + b(t) \geq I$ and, on $\mathcal{D}$,

$$\|(H' - \omega')v\| \leq c_1(t) \|K^{1/2} v\| + c_0(t) \|v\|$$

for a declared real differentiable scalar ω and integrable nonnegative c_0, c_1 . For each T define

$$\begin{aligned} E_T &= \int_0^T \|r\|, \quad E_{D,T} = \int_0^T \|r D_s\|, \quad B_T = \sup_{[0,T]} |b + \omega|, \\ C_T &= B_T E_T^2 + E_T E_{D,T}, \quad G_T = \int_0^T c_1, \\ A_T &= \sup_{0 \leq t \leq T} \left[\|r(t)\| + \|r(0)\| + \int_0^t \left(\|r' + i\omega r\| + c_0(s) \int_0^s \|r(u)\| du \right) ds \right]. \end{aligned}$$

When these quantities are finite, the causal solution belongs to the common strong domain and satisfies

$$\sup_{[0,T]} \|K^{1/2} \delta\| \leq \sqrt{E_T A_T + C_T} + \frac{1}{2} E_T G_T. \quad (8.4)$$

No spatial form norm of r and no application of H twice to δ is assumed.

Proof. We first justify the domain, rather than use a formal H^2 calculation. Set $B(t) = (\mathcal{L}(t) - i)^{-1}$. It maps the Hilbert–Schmidt space boundedly into its common graph domain and $B' = -B\mathcal{L}'B$. If $U(t, s)$ is the two-sided propagator, integration by parts in the graph norm gives

$$\begin{aligned} \delta(t) &= B(t)r(t) - U(t, 0)B(0)r(0) \\ &\quad + \int_0^t U(t, s) (Br + B\mathcal{L}'Br - Br') (s) ds. \end{aligned} \quad (8.5)$$

Indeed $\partial_s U(t, s) = iU(t, s)\mathcal{L}(s)$ and $i\mathcal{L}B = i - B$, so differentiating UBr recovers iUr minus the displayed integrand. This is precisely the Duhamel error. The terms are graph-integrable; absolutely continuous approximation in time proves the identity for the stated data.

Put $\eta = (\mathcal{L} - \omega)\delta - r$. Testing against common-domain adjoint solutions, or using (8.5), yields the mild identity

$$i\eta' = \mathcal{L}\eta - i(r' + i\omega r) + i(H' - \omega')\delta, \quad \eta(0) = -r(0).$$

It does not assert $\eta \in \mathcal{D}$. Let $\ell = \|(\mathcal{L} - \omega)\delta\|$, $e = \|\delta\|$ and $Q = \|K^{1/2}\delta\|$. Unitarity and the commuting right multiplication give $e(t) \leq E(t) = \int_0^t \|r\|$ and $\|\delta D_s\| \leq E_D(t) = \int_0^t \|r D_s\|$. The auxiliary equation and $K\delta = (\mathcal{L} - \omega)\delta + (b + \omega)\delta + \delta D_s$ therefore imply

$$\ell(t) \leq A_T + \int_0^t c_1 Q, \quad Q(t)^2 \leq E_T \ell(t) + C_T.$$

For $E_T > 0$, set $M(t) = A_T + \int_0^t c_1 Q$. Then $M' \leq c_1 \sqrt{E_T M + C_T}$, and differentiation of the square root, with a positive regularizer if necessary, gives $\sqrt{E_T M + C_T} \leq \sqrt{E_T A_T + C_T} + E_T \int_0^t c_1/2$. This proves (8.4). If $E_T = 0$, the causal error is identically zero. $\square$

The common-domain evolution premise is a standard nonautonomous evolution requirement; the regularity equivalence in [9] is relevant to checking it. The estimate retains the actual right-clock defect $E_{D,T}$ and the covariant temporal combination $r' + i\omega r$. A large internal operator is not a removable scalar phase.

For example, with $\Pi_j = p_j - e_j(A_{0,j} + a_j(t))$ in physical units and rate generator $H_{\text{kin}} = \sum_j \Pi_j^2 / (2M_j \hbar)$,

$$H'_{\text{kin}} = - \sum_j \frac{e_j}{M_j \hbar} a_{j,t} \cdot \Pi_j + i \sum_j \frac{e_j}{2M_j} \operatorname{div} a_{j,t}.$$

If K dominates the complete kinetic form, bounded vector additions give the admissible coefficients

$$c_1^2 = \sum_j \frac{2e_j^2 \|a_{j,t}\|_\infty^2}{M_j \hbar}, \quad c_0 = \sum_j \frac{|e_j| \|\operatorname{div} a_{j,t}\|_\infty}{2M_j} + \|(V_{\text{add}}/\hbar)' - \omega' I\|_\infty.$$

A linear force $F_j(t)x_j$ is also allowed when fixed positive traps provide a position form: Young's inequality supplies a shift $b \geq 1 + \|V_{\text{add}}^-\|/\hbar + \sum_j F_j^2 / (M_j \Omega_j^2 \hbar)$ and domination of $M_j \Omega_j^2 x_j^2 / (4\hbar)$. Its additional contribution to c_1^2 is $\sum_j 4|F_j'|^2 / (M_j \Omega_j^2 \hbar)$. A changing quadratic trap or an unbounded source-coordinate field needs its actual stronger graph bound.

8.4 Singular residuals and a safe common-domain alternative

Proposition 8.4 (Coulomb residual and temporal coordinates). *In three relative spatial dimensions, let ϕ be a smooth Gaussian nonzero at a collision. Then $|x|^{-1}\phi \in L_{\text{loc}}^2$ but $|x|^{-1}\phi \notin H_{\text{loc}}^1$.*

Nevertheless, if $V(x) = g/|x|$ is fixed in laboratory coordinates, $P_2(t, x)$ is a differentiable quadratic comparison potential and ϕ_t is a polynomial times a smooth finite Gaussian family, then

$$f = (V - P_2)\phi, \quad f_t = -P_{2,t}\phi + (V - P_2)\phi_t$$

are locally in L^2 and globally in L^2 for the indicated Gaussian polynomial families. On a finite parameter path with nondegenerate covariance they are continuous in L^2 . Under a homogeneous point change $x = L(t)X$, with invertible L ,

$$|\partial_t(g/|L(t)X|)| \leq \|L'L^{-1}\| |g|/|L(t)X|.$$

Proof. The radial square integrals near zero have orders $\int_0^1 dr$ for $1/r$ and $\int_0^1 r^{-2} dr$ for its leading gradient. The nonzero Gaussian value prevents cancellation of the latter leading term. Polynomial factors and Gaussian tails give the global L^2 statements; local domination by an integrable r^{-2} square envelope and uniform Gaussian tail bounds prove continuity. Finally differentiation gives $-g x \cdot (L'L^{-1}x)/|x|^3$, proving the last inequality. $\square$

An exponentially small nonzero Gaussian value does not remove the divergent spatial derivative. A translation that moves the collision point instead produces a $1/r^2$ temporal multiplier. The homogeneous repair retains the physical centroid, linear force, boost and scalar action; recentering them away may restore that moving singularity. These observations select a valid response method, without claiming a small apparatus error.

For completeness a strong-graph variant applies when a changing trap has a graph-relative rather than form-relative time derivative.

Proposition 8.5 (One graph by an ordered inverse lift). *Let $H(t)$ satisfy the common-domain assumptions above, and let $A = H + c \geq aI$ for constants $a > 0, c \geq 0$. Assume $\gamma(t) = \|H'(t)A(t)^{-1}\| \in L^1$. For $iE' = HE + f$, $E(0) \in \mathcal{D}$ and $f \in W^{1,1}([0, T]; \mathcal{H})$, set $R(t) = \int_0^t \gamma$. Then*

$$\begin{aligned} \|A(t)E(t)\| &\leq \|f(t)\| + e^{R(t)} \left[\|A(0)E(0)\| + \|f(0)\| \right. \\ &\quad \left. + \int_0^t e^{-R(s)} ((c + \gamma(s))\|f(s)\| + \|f'(s)\|) ds \right]. \end{aligned} \quad (8.6)$$

Proof. The auxiliary $Y = AE + f$ obeys $iY' = HY + cf + iH'E + if'$; the terms containing Hf cancel. Solve its Volterra equation using $E = A^{-1}(Y - f)$ and the bounded perturbation $H'A^{-1}$. The norm inequality for Y and Gronwall give (8.6). This is also a rigorous construction: $(A^{-1})' = -A^{-1}H'A^{-1}$ verifies the original equation weakly for the constructed E , and uniqueness of the Hilbert mild solution identifies it. No separate expression Hf is needed. $\square$

For a fixed laboratory Coulomb singularity and moving oscillator centres, the time derivative of the Coulomb kernel is zero. The derivative of the trap is quadratic and linear in the laboratory coordinates, so it can be graph-relative under the earned oscillator domain. Proposition 8.5, rather than an unjustified spatial derivative of f , is then the relevant interface. A detector moving with those centres still pays every term in (2.6).

9 A complete Coulomb forcing measure

This application keeps both the electron relative position and a quantum source coordinate. Its first calculation concerns a fixed Coulomb reference, whose exact spectral measure can be evaluated.

The second restores the actual source interaction and feedback. These are different estimates: a continuum-only coefficient of the fixed reference cannot remove the poles or replace the coupled resolvent of the actual parent.

Use reduced-mass atomic units and write

$$h = -\frac{1}{2}\Delta_r - \frac{1}{|r|}, \quad \phi(r) = \pi^{-1/2}e^{-|r|}, \quad f(r) = r_z\phi(r). \quad (9.1)$$

The Coulomb operator is self-adjoint on $H^2(\mathbb{R}^3)$ with form domain H^1 , ground energy $-1/2$ and normalized ground ϕ . Its excited bound energies are $E_n = -1/(2n^2)$, $n \geq 2$, with continuum $[0, \infty)$. We use the usual Coulomb spectrum, not a newly inferred set of levels. Only the isolated forcing f has the exact angular selection $l = 1, m = 0$; the actual coupled parent below retains every angular sector.

Theorem 9.1 (Full spectral measure of the dipole forcing). *The spectral measure of f under h is*

$$\mu_f = \sum_{n=2}^{\infty} d_n^2 \delta_{-1/(2n^2)} + 1_{E>0} \rho_C(E) dE, \quad (9.2)$$

where

$$d_n^2 = \frac{256n^7(n-1)^{2n-5}}{3(n+1)^{2n+5}}, \quad (9.3)$$

$$\rho_C(k^2/2) = \frac{256}{3} \frac{\exp[-4 \arctan(k)/k]}{(1+k^2)^5(1-e^{-2\pi/k})}, \quad k > 0. \quad (9.4)$$

Its total mass and first $(h+1)$ form moment are both one:

$$\int d\mu_f = 1, \quad \int E d\mu_f = 0, \quad \int (E+1) d\mu_f = 1. \quad (9.5)$$

On the whole true continuum,

$$\rho_C(E) \leq \rho_*, \quad (1+E)\rho_C(E) \leq \rho_*, \quad \rho_* = \frac{256}{3e^4}. \quad (9.6)$$

Proof. With $\psi = u(r)Y_{10}(\hat{r})/r$, the radial forcing and Friedrichs operator are

$$u_f = \frac{2}{\sqrt{3}}r^2e^{-r}, \quad h_1 = -\frac{1}{2}\partial_r^2 + r^{-2} - r^{-1}.$$

Integration of r^4e^{-2r} gives $\|f\|^2 = 1$. Direct differentiation gives $h_1u_f = (-1/2 + 1/r)u_f$, whence $\langle f, hf \rangle = 0$.

We specify the continuum normalization because an energy cross-section convention could otherwise introduce a wrong Jacobian. The regular Coulomb function is normalized by its unit-amplitude large-radius oscillation. In the standard convention [10],

$$\begin{aligned} u_k(r) &= \sqrt{2/\pi} F_1(-1/k, kr), \\ F_1(-1/k, kr) &= C_1(kr)^2 e^{-ikr} {}_1F_1(2 + i/k; 4; 2ikr), \\ C_1^2 &= \frac{2\pi(1+k^2)}{9k^3(1-e^{-2\pi/k})}. \end{aligned}$$

For $H_1^+ = G_1 + iF_1$, the Wronskian in r is $W_r(F_1, H_1^+) = -k$. The upper-boundary radial resolvent kernel is

$$(h_1 - E - i0)^{-1}(r, r') = \frac{2}{k} F_1(-1/k, kr_<) H_1^+(-1/k, kr_>). \quad (9.7)$$

Its derivative jump is -2 , as required by the coefficient $-1/2$ in h_1 . Its imaginary part is $(2/k)F_1(r)F_1(r')$. Stone's formula therefore gives the measure $u_k u_k dk$, or $u_k u_k/k$ per unit energy $E = k^2/2$. The Coulomb asymptotics used to choose H_1^+ are asymptotic statements at large radius, not exact finite-radius equalities.

Here is the overlap calculation. Put $a = 2 + i/k$, $s = 1 + ik$, $z = 2ik/(1 + ik)$. Laplace integration of the confluent hypergeometric series, first in its absolute-convergence region and then by analytic continuation, gives

$$\int_0^\infty r^4 e^{-sr} {}_1F_1(a; 4; 2ikr) dr = 24s^{-5} {}_2F_1(a, 5; 4; z).$$

The coefficient identity $(5)_j/(4)_j = 1 + j/4$ implies

$${}_2F_1(a, 5; 4; z) = (1 - z)^{-a} \left(1 + \frac{az}{4(1 - z)} \right).$$

The last bracket is $1/[2(1 - ik)]$. Consequently the integral is

$$\frac{12}{(1 + ik)^5(1 - ik)} \left(\frac{1 - ik}{1 + ik} \right)^{-2 - i/k}.$$

The continuous logarithm of the ratio is $-2i \arctan k$. Its contribution to the squared modulus is $e^{-4 \arctan(k)/k}$. Multiplication by the normalized radial prefactors gives $|\langle u_k, u_f \rangle|^2 = k \rho_C(k^2/2)$. The factor $1/k$ from Stone's energy measure proves (9.4).

The normalized negative-energy eigenfunction in this channel is

$$u_{n1}(r) = \frac{4r^2}{n^3} \sqrt{\frac{(n-2)!}{(n+1)!}} e^{-r/n} L_{n-2}^3(2r/n).$$

Insert $L_j^3 = (4)_j {}_1F_1(-j; 4; \cdot)/j!$ into the same Laplace identity. The bracket is now $n/[2(n-1)]$ and gives (9.3); in particular $d_2^2 = 32768/59049$.

For completeness, the radial regular/decaying solutions have only the simple negative poles $-1/(2n^2)$, $n \geq 2$. Their residues give precisely these normalized eigenprojections. On every compact positive-energy interval, (9.7) has continuous boundary values against the exponentially decaying forcing, so its forcing measure there is absolutely continuous. A remaining singular measure could only be supported at zero. The regular zero-energy solution is proportional to $\sqrt{r} J_3(\sqrt{8}r)$; its large-radius oscillatory magnitude is of order $r^{1/4}$ and it is not L^2 . Thus zero is not an eigenvalue and supplies no atom. A singular continuous measure cannot be supported on a single point. This proves completeness of (9.2). The already evaluated norm and first energy expectation now give (9.5); numerical quadrature is not used to establish them.

To prove the whole-continuum ceiling, let $A(k) = \arctan k - k/(1 + k^2)$. Since

$$\frac{d}{dk} \left\{ \frac{2k^3}{3(1 + k^2)} - A(k) \right\} = \frac{2k^4}{3(1 + k^2)^2} \geq 0,$$

we have $A(k) \leq 2k^3/[3(1 + k^2)]$. The logarithmic derivative of $(1 + k^2/2)\rho_C(k^2/2)$ is consequently at most

$$-\frac{16k}{3(1 + k^2)} + \frac{2\pi}{k^2(e^{2\pi/k} - 1)}.$$

For $k \leq 1$, use $e^x - 1 \geq x^3/6$ to bound the positive term by $3k/(2\pi^2)$; this leaves a strictly negative margin. For $k \geq 1$, use $e^x - 1 > x$ to bound it by $1/k$, whereas the negative magnitude is at least $8/(3k)$. The threshold limit is ρ_* . This proves the weighted ceiling and hence the unweighted one. $\square$

The bound lines are substantial. The formula gives $n^3 d_n^2 \rightarrow \rho_*$ and $E_{n+1} - E_n \sim n^{-3}$, matching the threshold density. For $n \geq 100$ one also has

$$d_n^2 \leq \frac{7}{4n^3}, \quad \sum_{n>N} d_n^2 \leq \frac{7}{8N^2}.$$

For example the logarithm of the factor multiplying $256/(3n^3)$ is at most $-4 + 10/n \leq -3.9$, and $256e^{-3.9}/3 < 7/4$ by a positive exponential Taylor sum. These tails can check a truncated calculation; they do not turn the discrete spectrum into a continuum.

9.1 A continuum-only coherent response

Let

$$H_s = \frac{\Omega}{2}(-\partial_Y^2 + Y^2), \quad U_s(t) = e^{-itH_s}, \quad \Omega > 0. \quad (9.8)$$

The source input is a vector in its physical-coordinate Hilbert space, possibly tensored with an inert finite internal reference.

Proposition 9.2 (Fixed-reference continuum response). *For the fixed parent $h + H_s$ and forcing $f \otimes a(t)$, let $q_C(0) = 0$ denote the electronic continuum response. Then*

$$\|q_C(t)\|^2, \quad \|(h+1)^{1/2}q_C(t)\|^2 \leq 2\pi\rho_* \int_0^t \|a(s)\|^2 ds, \quad (9.9)$$

$$\|H_s^{1/2}q_C(t)\|^2 \leq 2\pi\rho_* \int_0^t \|H_s^{1/2}a(s)\|^2 ds \quad (9.10)$$

whenever the displayed inputs are finite.

Proof. In the cyclic spectral representation generated by f , the continuum wave equals

$$q_C(E, t) = -iF(E)e^{-iEt}U_s(t) \int_0^t e^{iEs}U_s(-s)a(s) ds, \quad |F(E)|^2 = \rho_C(E).$$

Extend the time input by zero outside $[0, t]$. Hilbert-valued Plancherel, integrated first on the whole frequency line, gives the factor 2π for this unnormalized Fourier integral. Restricting to $E > 0$ and using (9.6) gives both estimates in (9.9). The source form commutes its own propagation, so the same argument applied to $H_s^{1/2}a$ gives (9.10). $\square$

For $a = gYp$, the last input is $gH_s^{1/2}Yp$, in that order. It is not $gYH_s^{1/2}p$. The source interaction picture here is used only for the norm proof; it has not changed the physical source coordinate or its current. The bound poles are excluded from this proposition and must be retained separately: a Poisson-smoothed atom contributes $d_n^2/(\pi\eta)$ at its center. There is no uniform η -independent full-spectrum density ceiling.

10 The actual compact-profile electron–source parent

Choose a real radial cutoff with $\chi_R = 1$ for $|r| \leq R$, $\chi_R = 0$ for $|r| \geq R + 1$, $0 \leq \chi_R \leq 1$ and $|\chi'_R| \leq 8$. Such a smooth flat cutoff is explicit: on $0 < t < 1$ use

$$\chi_R(R + t) = 1 - \frac{e^{-1/t}}{e^{-1/t} + e^{-1/(1-t)}}.$$

For $t \leq 1/2$ its derivative is bounded by $e^{2-1/t}(t^{-2} + 4) \leq 8$; reflection treats the other half. Set

$$d_R(r) = r_z \chi_R(|r|), \quad d_0 = \|d_R\|_\infty \leq R + 1, \quad H = h + H_s + gY d_R, \quad (10.1)$$

with real constant g . This is a prescribed compact-profile scalar parent. The finite annular electrostatic density $-\epsilon_0 \Delta d_R$ does not by itself construct its quantized source, field energy or laboratory realization.

Lemma 10.1 (Physical domains and forcing tail). *The parent (10.1) is self-adjoint on $D(h) \cap D(H_s)$ and has form domain $H^1(\mathbb{R}^3 \times \mathbb{R}) \cap \{Y\psi \in L^2\}$. It satisfies*

$$H + 2 + \frac{g^2 d_0^2}{\Omega} \geq I + \frac{1}{4} p_r^2 + \frac{1}{2} H_s. \quad (10.2)$$

For $f_R = d_R \phi$ and $R \geq 40$,

$$\|f_R - f\|^2 \leq \epsilon_R^2 := e^{-2R} \left(\frac{2}{3} R^4 + \frac{4}{3} R^3 + 2R^2 + 2R + 1 \right), \quad (10.3)$$

$$\|(h + 1)^{1/2} (f_R - f)\| \leq \sqrt{41.5 + R^{-2}} \epsilon_R. \quad (10.4)$$

At $R = 40$ these upper bounds are respectively $6 \cdot 10^{-15}$ in norm and $4 \cdot 10^{-14}$ in first-form norm.

Proof. Hardy and interpolation make Coulomb infinitesimally Laplacian bounded. The independent electron and oscillator operators commute; after a common positive shift their sum has domain $D(h) \cap D(H_s)$. Since $\|Y\psi\| \leq \sqrt{2/\Omega} \|H_s^{1/2} \psi\|$ and d_R is bounded, $gY d_R$ is infinitesimally operator bounded relative to this sum. This proves the operator assertion and the corresponding common closed form.

The square $\frac{1}{4} \|\nabla u + 2\hat{r}u\|^2 \geq 0$ gives $h + 1 \geq p_r^2/4$. Young's inequality gives $gY d_R \geq -\Omega Y^2/4 - g^2 d_0^2/\Omega$ and $H_s - \Omega Y^2/4 \geq H_s/2$, proving (10.2).

The omitted forcing is confined to $r \geq R$ and has magnitude at most that of u_f . Integrating $(4/3)r^4 e^{-2r}$ gives (10.3). For the radial error $w = (\chi_R - 1)u_f$, $|u'_f| \leq u_f$ on this tail and therefore $|w'| \leq 9u_f$. The positive upper bound on the remaining radial form potential is $1 + r^{-2}$. Thus

$$\langle w, (h_1 + 1)w \rangle \leq \left(\frac{81}{2} + 1 + R^{-2} \right) \int_R^\infty |u_f|^2 dr,$$

which is (10.4). Reciprocal positive Taylor sums for e^{80} enclose the stated numerical tails. $\square$

The cutoff is essential to the actual source parent. With an uncut $gY r_z$, completing the source square leaves $-g^2 r_z^2/(2\Omega)$. Electron packets translated to $r_z = L$ and source packets translated to $Y = -gL/\Omega$ have energy tending to $-\infty$. Thus no stable uncut source Hamiltonian is being approximated here. Equations (10.3)–(10.4) compare forcing vectors, not complete actual parents.

10.1 Exact blocks, including source feedback

Let

$$P = |\phi\rangle\langle\phi| \otimes I, \quad Q = I - P, \quad \Psi = \phi p + q, \quad q = Q\Psi.$$

The projection P does not reduce H . The odd profile has $Pd_RP = 0$, so the exact equations are

$$i\dot{q} = Aq + Lp, \quad A = QHQ, \quad Lp = gf_R \otimes Yp, \quad (10.5)$$

$$i\dot{p} = (H_s - \tfrac{1}{2})p + gY\langle\phi, d_Rq\rangle_r. \quad (10.6)$$

The subscript on the last inner product means integration in the electron coordinate only. The actual vector $p(Y, t)$ retains its source phase and every finite internal coefficient. It is not prescribed as a free source waveform.

On Q put

$$A_0 = Q(h + H_s)Q, \quad K_0 = A_0 + 1, \quad K = A + 1, \quad T_{\text{ph}} = I + \tfrac{1}{2}p_r^2 + H_s. \quad (10.7)$$

Here A_0 is the self-adjoint restriction of $h + H_s$ to $Q\mathcal{H}$. The spectral projection Q preserves the common operator and form domains. More precisely, A denotes the self-adjoint operator $A_0 + gYQd_RQ$ on $D(A_0)$: the electronic factor Qd_RQ is bounded and commutes with the source coordinate, so the same infinitesimal relative-bound argument used in Lemma 10.1 applies. Thus QHQ denotes this proved realization, rather than an assumed self-adjoint compression.

Since the complete excited electronic floor is $-1/8$,

$$K_0 \geq 7/8 + H_s, \quad K_0 \geq p_r^2/4 + H_s, \quad T_{\text{ph}} \leq (22/7)K_0. \quad (10.8)$$

For the last inequality take $4/11$ of the first lower bound and $7/11$ of the second, then multiply by $22/7$. This gives the exact coefficients 1 and $1/2$ of the identity and electron kinetic terms; the source coefficient is larger than required.

The unbounded source interaction has the bounded form sandwich

$$\|K_0^{-1/2}gYQd_RQK_0^{-1/2}\| \leq \theta := |g|d_0\sqrt{\frac{2}{\Omega(7/8)}}. \quad (10.9)$$

Indeed use $\|u\| \leq (7/8)^{-1/2}\|K_0^{1/2}u\|$ on one factor, and $\|Yv\| \leq \sqrt{2/\Omega}\|K_0^{1/2}v\|$ on the other. This bounds the full sesquilinear form and hence its operator. If $\theta < 1$, then

$$(1 - \theta)K_0 \leq K \leq (1 + \theta)K_0. \quad (10.10)$$

All these inequalities are physical form inequalities on the complete Q space. They neither truncate the oscillator nor delete QYd_RQ .

10.2 The spectral supremum and its correction

For $\zeta = E + i\eta$, $\eta > 0$, the spectral theorem gives

$$\begin{aligned} J_{\text{spec}}(E, \eta) &:= \|K_0^{1/2}(A_0 - \zeta)^{-1}K_0^{1/2}\| = \sup_{a \in \sigma(A_0)} \frac{a + 1}{|a - \zeta|} \\ &\leq \bar{J}(E, \eta) := \sup_{a \geq -1/8 + \Omega/2} \frac{a + 1}{|a - \zeta|}. \end{aligned} \quad (10.11)$$

The inequality can be strict. The half-line in the last expression contains spectral gaps and cannot generally replace $\sigma(A_0)$ in an equality.

Example 10.2 (A strict gap in the half-line bound). Take $\Omega = .01$, $E = -.117$, $\eta = .0001$. The lowest spectral points of A_0 are $-.120$ and $-.110$; the $n \geq 3$ electronic ladders begin above $-.051$, and the continuum begins at $.005$. At $a = E$, the half-line expression is 8830. On the actual spectrum the maximum is

$$J_{\text{spec}} = \frac{.88}{\sqrt{.003^2 + .0001^2}} < 294.$$

Indeed the first point gives a value above 280, while for every other spectral point $a \geq -.110$ the quotient is at most $.89/.007 < 128$. Thus the half-line equality fails by a large factor even for this concrete parent.

Theorem 10.3 (An earned low-window inverse for the actual block). *If $\bar{J}\theta < 1$, then*

$$\|K_0^{1/2}(A - \zeta)^{-1}LH_s^{-1/2}\| \leq \frac{|g|}{1 - \bar{J}\theta} \left\{ \frac{2}{\Omega} \sup_{s \geq \Omega/2} \int \frac{e + s + 1}{|e + s - \zeta|^2} \mu_{f_R}(de) \right\}^{1/2}. \quad (10.12)$$

For the static row

$$\Omega = 1/100, \quad |g| \leq 10^{-6}, \quad R = 40, \quad d_0 \leq 41, \quad E \leq -.47, \quad \eta > 0, \quad (10.13)$$

the complete physical sandwich satisfies

$$\|T_{\text{ph}}^{1/2}(A - E - i\eta)^{-1}LH_s^{-1/2}\|^2 < 4.035635 \cdot 10^{-9} < 4.05 \cdot 10^{-9}. \quad (10.14)$$

Proof. Let $V_0 = K_0^{-1/2}gYQd_RQK_0^{-1/2}$ and $B_0 = K_0^{1/2}(A_0 - \zeta)^{-1}K_0^{1/2}$. The bounded form pencil of $A - \zeta$ has inverse

$$K_0^{1/2}(A - \zeta)^{-1}K_0^{1/2} = (I + B_0V_0)^{-1}B_0.$$

The order is fixed: the pencil is $B_0^{-1} + V_0 = B_0^{-1}(I + B_0V_0)$. The Neumann bound is $\|(I + B_0V_0)^{-1}\| \leq (1 - \bar{J}\theta)^{-1}$. This is the concrete form-inverse construction of Theorem 5.1; no Hilbert-norm boundedness of YQd_RQ has been assumed.

In the baseline squared norm, integrate out only the electron spectral measure. The remaining positive source operator is

$$H_s^{-1/2}YF(H_s)YH_s^{-1/2}, \quad F(s) = \int \frac{e + s + 1}{|e + s - \zeta|^2} \mu_{f_R}(de).$$

Bound F by its supremum, then use $\|YH_s^{-1/2}\| \leq \sqrt{2/\Omega}$. This proves (10.12). Moving Y through $F(H_s)$ or through the resolvent would not prove that estimate.

For (10.13), $\theta < .00062$ and $\bar{J} \leq 88/35$. To see the latter, first lower every denominator to $a + .47$ and use $a \geq -.12$; the resulting $(a + 1)/(a + .47)$ decreases and has value $88/35$ at the floor. Thus the correction in (10.11) leaves the low-window bound unchanged.

The source supremum also needs its own argument. First dominate the integral kernel at the fixed endpoint $E = -.47, \eta = 0$, since $e + s \geq -.12$. At this endpoint the derivative of $(e + s + 1)/(e + s + .47)^2$ with respect to s has sign $-(e + s + 1.53) < 0$. It is consequently enough to use $s = 1/200$. This monotonicity was not asserted for every negative E before making the endpoint domination.

For the uncut forcing, isolate the exact $2p$ mass $w_2 = 32768/59049$. All remaining spectral mass lies at $e \geq -1/18$, and the same endpoint kernel is decreasing in e . Its complete bound-plus-continuum integral is therefore

$$F_f(1/200) \leq w_2 \frac{352}{49} + (1 - w_2) \frac{123048}{22801} < 6.4. \quad (10.15)$$

This analytic ceiling requires neither a continuum quadrature nor a truncation of the Rydberg series.

The compact forcing changes the baseline weighted inverse norm, before multiplication by $|g|$, by at most

$$\bar{J} \epsilon_R \sqrt{\frac{2}{\Omega(7/8)}}.$$

This follows by inserting $K_0^{-1/2}$ and applying its lower form floor to $f_R - f$ and the ordered source factor. Use (10.8), $\epsilon_R < 6 \cdot 10^{-15}$, $\sqrt{1280} < 35.778$ and $\sqrt{1600/7} < 16$. The resulting entirely rational upper bound is

$$\frac{22}{7} 10^{-12} \frac{[35.778 + (88/35)(6 \cdot 10^{-15})16]^2}{[1 - (88/35)(.00062)]^2} < 4.035635 \cdot 10^{-9}.$$

This proves (10.14), retaining the actual excited-block feedback in the inverse. $\square$

The result is a uniform frequency-window statement with source weight $H_s^{1/2}$. It does not say that the actual time-dependent input p has Fourier support in this window. Such a use requires the temporal band and complementary-frequency terms of the window theorem. We next give a different all-frequency bound with the stronger, explicitly earned source weight H_s .

11 Actual source graphs and finite-time complete currents

Write $m_0 = \|f_R\|^2$ and $m_1 = \|(h+1)^{1/2}f_R\|^2$. The form tail gives $m_0 \leq 1$ and, at $R = 40$, $m_1 < (1 + 4 \cdot 10^{-14})^2 < 1 + 10^{-12}$. The source ladder operators give the ordered bounds

$$\|YH_s^{-1}\| \leq \frac{4\sqrt{2}}{3\Omega}, \quad \|H_s^{1/2}YH_s^{-1}\| \leq \frac{\sqrt{3} + 1/\sqrt{2}}{\sqrt{\Omega}}. \quad (11.1)$$

Indeed $H_s = \Omega(N + 1/2)$ and $Y = (a + a^\dagger)/\sqrt{2}$. The downward and upward coefficients of aH_s^{-1} and $a^\dagger H_s^{-1}$ are bounded by $2/(3\Omega)$ and $2/\Omega$, respectively. For the weighted versions, their squared ratios are

$$\frac{n(n-1/2)}{\Omega(n+1/2)^2} \leq \frac{1}{\Omega}, \quad \frac{(n+1)(n+3/2)}{\Omega(n+1/2)^2} \leq \frac{6}{\Omega}.$$

The first inequality is direct and the second ratio decreases from its value 6 at $n = 0$. Their sum in Y proves (11.1). These are full-source operator bounds, not a bound on a selected actual occupation.

Consequently

$$\|K_0^{1/2}(f_R \otimes Y)H_s^{-1}\|^2 \leq C_\Omega^2 := \frac{32m_1}{9\Omega^2} + \frac{(\sqrt{3} + 1/\sqrt{2})^2 m_0}{\Omega}. \quad (11.2)$$

The two terms follow by splitting $K_0 = (h+1) + H_s$ on the forcing vector, with the ordering in (11.1) preserved.

Proposition 11.1 (All-frequency response with a physical source graph). *For $\theta < 1$ and every $E \in \mathbb{R}$, $\eta > 0$,*

$$\|T_{\text{ph}}^{1/2}(A - E - i\eta)^{-1}LH_s^{-1}\|^2 \leq \frac{22}{7} \frac{1 + \theta}{1 - \theta} \frac{g^2 C_\Omega^2}{\eta^2}. \quad (11.3)$$

For the actual input p of (10.6), with $q(0) = 0$, let q_T agree with its forced Q equation on $[0, T]$ and then evolve homogeneously with A . If $H_s p \in L^2(0, T)$, then

$$\int_0^\infty e^{-2\eta t} \|T_{\text{ph}}^{1/2} q_T(t)\|^2 dt \leq \frac{22}{7} \frac{1 + \theta}{1 - \theta} \frac{g^2 C_\Omega^2}{\eta^2} \int_0^T e^{-2\eta s} \|H_s p(s)\|^2 ds. \quad (11.4)$$

Also, whenever $H_s p \in L^1(0, t)$,

$$Q(t) := \|T_{\text{ph}}^{1/2} q(t)\| \leq |g| \sqrt{\frac{22}{7} \frac{1+\theta}{1-\theta}} C_\Omega \int_0^t \|H_s p(s)\| ds. \quad (11.5)$$

Proof. By (10.8) and (10.10), $T_{\text{ph}} \leq 22K/[7(1-\theta)]$. The form $K = A + 1$ commutes with the actual A resolvent, whose unweighted norm is at most $1/\eta$. Its input satisfies $\|K^{1/2} L H_s^{-1}\| \leq \sqrt{1+\theta} |g| C_\Omega$. Their ordered product proves (11.3). The finite lower bound on K is positive; rescaling by that bound gives the equivalent unit-coercive form convention if needed. No commutation of T_{ph} with A is used.

Extend $1_{[0,T]} H_s p$ by zero. The causal exponentially damped Fourier transform of the Duhamel equation is the resolvent multiplier applied to this complete source vector. Plancherel with (11.3) proves (11.4), as in Theorem 6.1. There is no derivative of the time cutoff in this argument and no frequency-band assumption. Finally Duhamel in the stationary positive K form and $\|K^{1/2} e^{-itA} v\| = \|K^{1/2} v\|$ give (11.5). $\square$

The $1/\eta^2$ cost includes all actual bound poles and continuum. It cannot be discarded because the fixed-reference continuum-only calculation happened to be η independent. The source weights in (10.12) and (11.3) are also different.

Proposition 11.2 (Source graphs for the actual feedback evolution). *Let the normalized entrance be $\Psi(0) = \phi \otimes |1\rangle \otimes \zeta$, with any normalized passive finite internal vector ζ . Under the actual static parent (10.1), put $a = |g|d_0$. Then*

$$H_1(t) := \|H_s^{1/2} \Psi(t)\| \leq \sqrt{3\Omega/2 + a\sqrt{\Omega/2} t}, \quad (11.6)$$

$$\|H_s \Psi(t)\| \leq \Omega\{3/2 + \sqrt{3}at + a^2 t^2/2\}, \quad (11.7)$$

$$e(t) := \|q(t)\| \leq |g|\{\sqrt{3}t + at^2/2\}. \quad (11.8)$$

Since P commutes with H_s , the first two bounds also control the corresponding graphs of the actual p .

Proof. The commutator is $[H_s, H] = -ig\Omega d_R p_Y$, with $p_Y = -i\partial_Y$. The expectation identity and $\|p_Y \Psi\| \leq \sqrt{2/\Omega} \|H_s^{1/2} \Psi\|$ give

$$\left| \frac{d}{dt} H_1^2 \right| \leq a\sqrt{2\Omega} H_1.$$

Regularizing the square root if necessary and integrating gives (11.6). The norm derivative for $H_s \Psi$ obeys

$$\frac{d}{dt} \|H_s \Psi\| \leq \|[H_s, H] \Psi\| \leq a\sqrt{2\Omega} H_1(t).$$

Its initial norm is $3\Omega/2$; integrating the preceding linear bound gives (11.7).

These computations can be justified without assuming the future source stays Gaussian. Let P_N project onto source occupations $0, \dots, N-1$, and use the full-space operators

$$H_N = h + H_s + g d_R P_N Y P_N.$$

They are self-adjoint on $D(h) \cap D(H_s)$ with uniform infinitesimal relative bounds and the coercivity (10.2). For every vector in this common domain, $H_N u \rightarrow H u$, because P_N commutes with H_s and converges in its form norm. The resolvent identity, with the uniformly bounded nonreal resolvents on its left, gives strong resolvent convergence; the associated unitary groups converge strongly on

compact time intervals. For $N \geq 2$ the entrance lies in $P_N \mathcal{H}$ and remains there, so both source graphs are bounded operators on that invariant subspace. The commutator is $-ig\Omega d_R P_N p_Y P_N$ and obeys the same uniform estimates as above. Hence the displayed graph bounds hold for $e^{-itH_N} \Psi(0)$. At each time, strong wave convergence and weak lower semicontinuity of the closed graphs of $H_s^{1/2}$ and H_s pass these bounds to the actual wave. No finite source truncation remains in the result.

Lastly, from the exact equation for q and its zero entrance,

$$\|q(t)\| \leq |g| \|f_R\| \int_0^t \|Yp(s)\| ds \leq |g| \sqrt{2/\Omega} \int_0^t H_1(s) ds.$$

Here $\|f_R\| \leq 1$ and P commutes with H_s . This is (11.8). The feedback term in (10.6) has not been set to zero anywhere. $\square$

11.1 All coherent terms in the original physical currents

The nominated current constitution is the nonrelativistic canonical one:

$$j_r[\Psi] = \text{Im } \Psi^\dagger \nabla_r \Psi, \quad j_Y[\Psi] = \Omega \text{Im } \Psi^\dagger \partial_Y \Psi.$$

Define differences relative to the *actual* P wave,

$$\delta\rho = |\Psi|^2 - |\phi p|^2, \quad \delta j = j[\Psi] - j[\phi p].$$

The vector ϕp is generally unnormalized and does not solve an autonomous closed evolution. It is a wave comparison, not a separately assigned physical flow. To make its continuity defect explicit, set $\mathcal{F} = gY \langle \phi, d_R q \rangle_r$ and $M(t) = \|p(t)\|^2$. The feedback equation gives, in distributions,

$$\partial_t |\phi p|^2 + \text{div } j[\phi p] = s_P := 2|\phi|^2 \text{Im}(p^\dagger \mathcal{F}), \quad M = 1 - \|q\|^2, \quad M' = \int s_P. \quad (11.9)$$

The products are integrable under the earned source graphs: for example $\|\mathcal{F}\| \leq |g| d_0 \|Yq\|$. Even if $M > 0$ and the comparison is normalized, writing $\rho_P = |\phi p|^2$, its density ρ_P/M and current $j[\phi p]/M$ have source $s_P/M - M' \rho_P/M^2$. Normalization therefore does not create conservative equivariance. The estimates below retain the original unnormalized comparison and need neither a second flow nor a continuity-defect allowance.

Lemma 11.3 (Full electron and source current differences). *With $d_r = \|\nabla_r q\|$ and $d_Y = \|\partial_Y q\|$,*

$$\|\delta\rho\|_1 \leq 2e + e^2, \quad (11.10)$$

$$\|\delta j_r\|_1 \leq d_r + e + e d_r, \quad (11.11)$$

$$\|\delta j_Y\|_1 \leq \Omega \{(1+e)d_Y + e\sqrt{2/\Omega} H_1(t)\}, \quad (11.12)$$

where $d_r \leq \sqrt{2} Q(t)$ and $d_Y \leq \sqrt{2/\Omega} Q(t)$.

Proof. For any derivative D , expand before estimating:

$$j_D[\phi p + q] - j_D[\phi p] = c_D \text{Im}\{(\phi p)^\dagger Dq + q^\dagger D(\phi p) + q^\dagger Dq\}.$$

Thus every interference term and the quadratic error current remain. Cauchy–Schwarz, $\|p\| \leq 1$ and $\|\nabla\phi\| = 1$ give the electron estimate. For the source, $\|\partial_Y p\| \leq \sqrt{2/\Omega} H_1$ because P commutes with the source kinetic form. Multiplication by $c_Y = \Omega$ gives (11.12). The density expansion gives (11.10). Finally the electron and source kinetic terms are included with their stated coefficients in T_{ph} , giving the two derivative bounds. Spectral orthogonality of P and Q has not been used to discard a pointwise cross term. $\square$

Corollary 11.4 (An explicit finite-time current bound). *For the same static parent with $\Omega = .01$, $g = 10^{-6}$, $R = 40$, $d_0 \leq 41$ and the entrance of Proposition 11.2,*

$$Q(100) < .00051, \quad e(100) < .000176, \quad (11.13)$$

$$\int_0^{100} \|\delta j_r(t)\|_1 dt < .045033131 < .046, \quad (11.14)$$

$$\int_0^{100} \|\delta j_Y(t)\|_1 dt < .003970830 < .004. \quad (11.15)$$

Proof. Equations (10.4), (10.9) and (11.2) give the rational enclosure

$$\frac{22}{7} \frac{1 + .00062}{1 - .00062} \left\{ \frac{32(1 + 10^{-12})}{9(.01)^2} + \frac{25}{4(.01)} \right\} < 338^2.$$

Here $(\sqrt{3} + 1/\sqrt{2})^2 = 7/2 + \sqrt{6} < 25/4$. Let $a = 41 \cdot 10^{-6}$. Equations (11.5)–(11.8) consequently admit the polynomial upper bounds

$$\begin{aligned} \overline{Q}(t) &= 338 \cdot 10^{-8} \{ (3/2)t + (7/8)at^2 + a^2t^3/6 \}, \\ \overline{e}(t) &= 10^{-6} \{ (7/4)t + at^2/2 \}, \\ \overline{H}_1(t) &= .123 + .071at. \end{aligned}$$

All roundings are outward: $\sqrt{3} < 7/4$, $\sqrt{.015} < .123$ and $\sqrt{.005} < .071$. Substitute $d_r \leq (10/7)\overline{Q}$ and $d_Y \leq 15\overline{Q}$, using $\sqrt{2} < 10/7$, $\sqrt{200} < 15$. The two current integrands are bounded by

$$\frac{10}{7}\overline{Q} + \overline{e} + \frac{10}{7}\overline{e}\overline{Q}, \quad .01\{15(1 + \overline{e})\overline{Q} + 15\overline{e}\overline{H}_1\}.$$

These are explicit nonnegative rational polynomials. Exact integration on $[0, 100]$ gives respectively $0.045033130836\dots$ and $0.003970829749\dots$. Direct evaluation of the same polynomials gives (11.13). The exact integrals are

$$\frac{113483489708747242649}{252000000000000000000}, \quad \frac{95299913979542242649}{2400000000000000000000},$$

so integer comparison verifies both printed outward ceilings. No continuum quadrature enters this bound. $\square$

The electron integral has units of relative-coordinate length, and the source integral of source-coordinate length in the chosen dimensionless Y convention. They bound $j[\Psi] - j[\phi p]$, not the entire absolute currents. They are not event probabilities. A detector or tube claim requires a corresponding surface/guard and an original-law transport interface with its own constants. Comparing to a freely propagated intended source instead of the actual p would additionally require the feedback phase and current cost of $p - p_{\text{free}}$.

The compact scalar and oscillator constitute a definite mathematical example with actual source feedback, full Coulomb poles and continuum, and complete canonical currents. A finite-mass source/COM reduction, source field energy, retardation, loading, clock and a microscopic Pauli or Dirac current identification remain separate physical questions. None is inferred from the exact Coulomb calibration or from the small forcing tail.

11.2 Flow and companion-paper interfaces

For the entrance of Proposition 11.2, the actual parent (10.1) is the constant- g case of the *Full Coulomb–source flow* theorem in the revised companion paper [3]. The reduced-mass units, the smooth compact dipole and its full collar, the first excited source entrance, the passive internal factor, and both canonical currents agree. Constant g satisfies that theorem’s bounded absolutely continuous control hypothesis on every finite horizon, including $[0, 100]$. Its common first-domain, tangent-regularity and logarithmic-current arguments consequently supply a deterministic reference-compatible flow for this actual wave, with reference-almost-sure collision and node avoidance. An original joint law absolutely continuous with respect to the entrance wave density is transported by reweighting those same paths once. A quantitative domination factor for (2.7) is a further premise, not a consequence of absolute continuity.

The current bounds above do not apply that flow theorem to ϕp . Conservative two-flow stability in the revised P3 requires two admitted conservative flows, a common positive tube, and a bound for the probability of leaving it. Its separate signed-source appendix requires a normalized reference, classical local regularity, and finite source and divergence costs. Neither conclusion follows by analogy from (11.9). No variable-coefficient spin-curl current is used here; adding one would require its own derivative regularity and current estimates.

The Gaussian preparation and instrument results of P1 [1] and the finite repeated-record protocol of P2 [2] concern their own effective parents, complete entrance laws and decoders. The present estimates establish neither those entrance premises nor fresh independence, renewed readiness, or repeated unknown-input measurement. Conversely, wave reset or preparation of an actual law does not supply a missing derivative, source graph or flow hypothesis for the present parent. Any composition must verify the same complete configuration and every retained correlation on the same time interval.

12 Significance and remaining scope

The results give explicit routes from a complete coherent forcing to a physical derivative form and then to every admitted coordinate current. The inverse-lift argument handles a form-dual residual through its temporal regularity. The window argument handles changing input directions without discarding the operator response. The ordered resolvent formulas retain source interactions and physical weights without a false commutation. The Coulomb example discharges these questions in a definite electron–oscillator parent, including the source graph and the feedback-modified input.

The distinctions in the hypotheses are productive. They identify whether a calculation still needs a residual derivative, a physical form floor, a spectral coefficient, endpoint dressing, a source moment, or a common-domain propagation argument. They also expose specific invalid substitutions: a tiny collision tail does not make a Coulomb residual differentiable in space; a source-only truncation does not suppress propagated high-sector response; a reference invariant is not the physical current metric; and a shifted spectral floor is not an actual spectral gap.

A physical record application must still supply its original wave and actual law, the complete Hamiltonian/current constitution, an earned flow, a decoder and a whole retention interval. The current estimates can then enter its guard or moving-surface argument with the relevant length scales and original-law factor. No common apparatus joining all four portfolio papers has been inferred. Finite material sources, microscopic current reductions, preparation warrant and attainable control errors remain separate research questions.

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AI assistance and verification. AI systems assisted with mathematical drafting, literature comparison, algebraic and numerical checks, and manuscript preparation. The author remains responsible for the arguments and presentation. This assistance does not constitute independent scientific validation.

Independent review and collaboration. The author welcomes independent mathematical review of the common-form and domain arguments, computational replication of the explicit spectral and current bounds, and physical collaboration on complete source/control models and faithful retained records.

A Spatial forcing compression with all sources retained

A bounded relative-position region can have discrete free spatial modes even when the complete source Hilbert space has continuous spectrum. The following construction truncates the spatial forcing directions and leaves every source coefficient exact.

Theorem A.1 (Full-source inverse defect). *Let $\mathcal{H} = L^2(\Omega; \mathcal{Z})$, where $\mathcal{Z}$ includes all source, centre-of-mass and internal variables. Let a be a closed positive form with operator K and*

$$a(v, v) \geq \|v\|^2 + \lambda \|\nabla_{\text{rel}} v\|^2 + \text{the required hidden kinetic forms.}$$

Choose a bounded open relative-position domain $D \subset \Omega$, with free Dirichlet eigenvalues μ_j , and project onto its first N spatial eigenfunctions tensored with $I_{\mathcal{Z}}$. In the full physical form domain let $\mathcal{W}_N$ consist of zero extensions of vectors in $H_0^1(D; \mathcal{Z})$ whose coefficients in those N modes vanish. This defines the zero Dirichlet condition without requiring a classical boundary trace. Let $\mathcal{V}_N = \mathcal{W}_N^{\perp a}$ and let K_N be the restricted closed form operator on the Hilbert closure of $\mathcal{V}_N$. With its physical inclusion J_N , define

$$B = K^{-1}, \quad B_N = J_N K_N^{-1} J_N^\dagger, \quad D_N = B - B_N.$$

Then $D_N \geq 0$, its range is in $\mathcal{W}_N$, and

$$\|D_N\| \leq \delta_N, \quad \|K^{1/2} D_N\| \leq \sqrt{\delta_N}, \quad \delta_N = (1 + \lambda \mu_{N+1})^{-1}. \quad (\text{A.1})$$

All exterior responses and Dirichlet form traces of Bh and $B_N h$ coincide. If $D \subset [-a, a]^3$ and $N = n^3$, a sufficient bound is $\delta_N \leq [1 + \lambda \pi^2(n+1)^2/(4a^2)]^{-1}$.

Proof. The coercive gradient bound makes form convergence imply convergence in the relative H^1 norm. The zero extensions of $H_0^1(D; \mathcal{Z})$ form a closed subspace there; the spatial-mode coefficient conditions are closed already in L^2 . Thus $\mathcal{W}_N$ and its form orthogonal complement are closed. Decompose the Riesz solution $u = Bh$ into $u_N + w_N$. Testing its form equation in each orthogonal subspace gives $u_N = B_N h$, $w_N = D_N h$ and

$$a(D_N h, D_N h) = \langle h, D_N h \rangle.$$

In particular D_N is the positive inverse associated with the restricted high form, included into the physical space. The vector spectral expansion in the free spatial basis gives $\|\nabla_{\text{rel}} w\|^2 \geq \mu_{N+1} \|w\|^2$ on $\mathcal{W}_N$. Coercivity and Cauchy–Schwarz therefore yield $\|D_N h\| \leq \delta_N \|h\|$; substitution in the last identity proves the form bound. Its support and zero Dirichlet trace give the exterior claim; equality of conormal derivatives is not asserted. Dirichlet domain monotonicity compares D to the cube. Any cube mode below $\pi^2(n+1)^2/(4a^2)$ has all three indices at most n , so there are at most n^3 such modes, proving the stated ceiling. $\square$

If the high form is dense in the corresponding high Hilbert space, D_N is its compressed-form inverse, included by zero extension. Otherwise its Hilbert closure is used; the proof is unchanged. It is generally *not* the unrestricted inverse followed by a high projection. For example, with $K = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ and $\mathcal{W} = \text{span}(0, 1)$, $D = \text{diag}(0, 1/2)$, whereas $K^{-1}Q = \begin{pmatrix} 0 & -1/3 \\ 0 & 2/3 \end{pmatrix}$ is not even self-adjoint.

The decreasing high spaces have zero intersection by completeness of the free spatial expansion. Their orthogonal projections in the form Hilbert space converge strongly to zero: nested projection differences have squared norms equal to the differences of the squared projection norms, and the limit lies in the intersection. Hence the retained spaces are form dense. One may enrich them by any finite list of original form columns by intersecting the high space with their form orthogonal complements. The defect decreases and (A.1) survives; an enriched $f \in \text{Dom } K$ satisfies $B_N K f = f$. No actual population has been substituted.

The construction supplies an inverse error and exact source retention, not a finite apparatus simulation: each spatial coefficient remains in $\mathcal{Z}$, and boundary response spaces can be infinite dimensional. Any evolution or control estimate must still propagate the physical form under its own hypotheses. Once that estimate supplies e, Q , Proposition 2.1 controls every admitted coordinate current, including source currents.

B A small omitted source does not remove causal feedback

Proposition B.1 (Retained high response and endpoint dressing). *For an admitted same-parent block evolution, suppose*

$$iu'_L = A_L u_L + B^\dagger u_H + r_L, \quad iu'_H = Du_H + Bu_L + r_H.$$

Then, with the true high-block propagator U_H ,

$$u_H(t) = U_H(t, 0)u_H(0) - i \int_0^t U_H(t, s)(B(s)u_L(s) + r_H(s)) ds.$$

Thus the low equation retains an entrance term, the high source, and the retarded feedback kernel $-iB(t)^\dagger \int_0^t U_H(t, s)B(s)u_L(s) ds$. For a fixed $D \geq gI > 0$ and absolutely continuous $F = Bu_L + r_H$,

$$\begin{aligned} u_H(t) &= e^{-itD}u_H(0) - D^{-1}F(t) + e^{-itD}D^{-1}F(0) \\ &\quad + D^{-1} \int_0^t e^{-i(t-s)D}F'(s) ds. \end{aligned} \tag{B.1}$$

The last three terms have $D^{1/2}$ norm at most $g^{-1/2}(\|F(t)\| + \|F(0)\| + \int_0^t \|F'\|)$.

Proof. The first formula is variation of constants in the true high block. Substitution into the low equation gives the kernel and both other terms. Since $\partial_s e^{-i(t-s)D} = iDe^{-i(t-s)D}$, integration by parts gives (B.1); the spectral theorem bounds $\|D^{-1/2}\| \leq g^{-1/2}$. $\square$

After removal of a constant real scalar frequency ω , the relevant inverse is $(D - \omega)^{-1}$ and the temporal quantity is $F' + i\omega F$. A retained right clock must also be restored in that combination. An angular kinetic gap in a frozen model does not by itself prove the required gap of a full radial/source block.

The elementary two-level parent $H = \begin{pmatrix} 0 & b \\ b & \Delta \end{pmatrix}$ already isolates the issue. With original state $(1, 0)$ and no forcing, the high population is

$$\frac{4b^2}{\Delta^2 + 4b^2} \sin^2\left(\frac{1}{2}t\sqrt{\Delta^2 + 4b^2}\right).$$

This follows by subtracting $\Delta/2$ times the identity and squaring the remaining traceless matrix. An exactly zero high source therefore does not imply a zero high response. For a forced zero-entrance solution with source $(g_0, 0)$ one instead has $u_H''(0) = -bg_0$, a separate causal counterexample.

In angular tensor applications a rank-two coupling connects both top bands $l = L - 1, L$ to omitted $L + 1, L + 2$ unless a genuine parity restriction has been established. A concrete molecular application must identify its full tensor coupling, correlated entrance and local forcing bounds. Small forcing bounds do not replace Bu_L , its propagated angular/source derivatives, or the original entrance error in the formulas above. Coherent currents can respond at first order in a high amplitude while its population is second order: the current conversion must retain the cross terms even when the high population is small.

References

- [1] J. Rodgers, *Conditional Gaussian Preparation with Retained Archives*, companion manuscript (2026). [doi:10.5281/zenodo.23259560](https://doi.org/10.5281/zenodo.23259560).
- [2] J. Rodgers, *Effective Repeated Position Records with Retained Entropy and Calibrated Reset*, companion manuscript (2026). [doi:10.5281/zenodo.23259558](https://doi.org/10.5281/zenodo.23259558).
- [3] J. Rodgers, *Reference-Weighted Deterministic Quantum Flows with Singular Interactions and Retained Sources*, revised companion manuscript (2026). [doi:10.5281/zenodo.23259569](https://doi.org/10.5281/zenodo.23259569).
- [4] J. Rodgers, *Autonomous Quantum Measurement Chains with Faithful Equilibrium Records*, consolidated revision of the version 2 massive-configuration preprint, 4 October 2026. [doi:10.5281/zenodo.23131069](https://doi.org/10.5281/zenodo.23131069).
- [5] J. Rodgers, *Nonequilibrium Calibration and Faithful Records in Autonomous Effective Measurement Models*, version 2, 4 October 2026 manuscript. [doi:10.5281/zenodo.23131081](https://doi.org/10.5281/zenodo.23131081).
- [6] T. Kato, Wave operators and similarity for some non-selfadjoint operators, *Mathematische Annalen* **162** (1966), 258–279. [doi:10.1007/BF01360915](https://doi.org/10.1007/BF01360915).
- [7] P. D’Ancona, Kato smoothing and Strichartz estimates for wave equations with magnetic potentials, *Communications in Mathematical Physics* **335** (2015), 1–16. [doi:10.1007/s00220-014-2169-8](https://doi.org/10.1007/s00220-014-2169-8); [arXiv:1403.2537](https://arxiv.org/abs/1403.2537).
- [8] J. Kiszyński, Sur les opérateurs de Green des problèmes de Cauchy abstraits, *Studia Mathematica* **23** (1964), 285–328. [doi:10.4064/sm-23-3-285-328](https://doi.org/10.4064/sm-23-3-285-328).
- [9] J. Schmid and M. Griesemer, Kato’s theorem on the integration of non-autonomous linear evolution equations, *Mathematical Physics, Analysis and Geometry* **17** (2014), 265–271. [doi:10.1007/s11040-014-9154-5](https://doi.org/10.1007/s11040-014-9154-5); [arXiv:1203.4700](https://arxiv.org/abs/1203.4700).
- [10] National Institute of Standards and Technology, *Digital Library of Mathematical Functions*, [Section 33.2](#) and [Section 33.11](#), Coulomb functions. Accessed 9 October 2026.
- [11] G. Panati, H. Spohn and S. Teufel, Space-adiabatic perturbation theory, *Advances in Theoretical and Mathematical Physics* **7** (2003), 145–204. [doi:10.4310/ATMP.2003.v7.n1.a6](https://doi.org/10.4310/ATMP.2003.v7.n1.a6); [arXiv:math-ph/0201055](https://arxiv.org/abs/math-ph/0201055).
- [12] H. R. Lewis, Jr., Classical and quantum systems with time-dependent harmonic-oscillator-type Hamiltonians, *Physical Review Letters* **18** (1967), 510–512. [doi:10.1103/PhysRevLett.18.510](https://doi.org/10.1103/PhysRevLett.18.510); erratum, *Physical Review Letters* **18** (1967), 636, [doi:10.1103/PhysRevLett.18.636.2](https://doi.org/10.1103/PhysRevLett.18.636.2).
- [13] H. R. Lewis, Jr. and W. B. Riesenfeld, An exact quantum theory of the time-dependent harmonic oscillator and of a charged particle in a time-dependent electromagnetic field, *Journal of Mathematical Physics* **10** (1969), 1458–1473. [doi:10.1063/1.1664991](https://doi.org/10.1063/1.1664991).